

MR Known Artifacts



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Competencies

After completing this session, you will be able to:

- Recognize a specific artifact
- Identify the problems associated with this artifact
- Identify and recognize if the artifact is a hardware or an application related issue
- Know how to solve it

Applications Related Artifacts

Annefact

Startifact

Maxwell Term Artifact

Gibbs Artifact

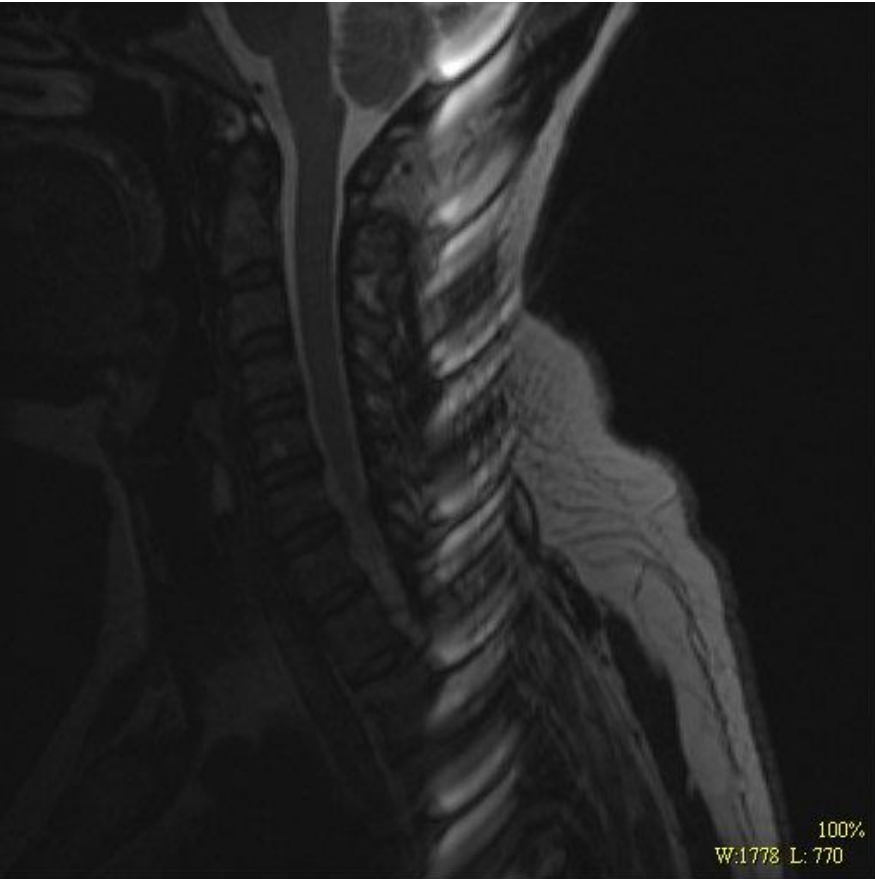
Fineline Artifact

Twinspeed Zoom Artifacts

Respiratory Motion Artifacts

Phase Wrap (Aliasing)

Annefact



Issue description:

Annefact which is common with a Sagittal Spine scan with the CTL coil.

Solution:

Customer needs to use the recommended methods of coil selection and field of view size to minimize the occurrence of this artifact. In addition, position the center of the FOV (S-I direction) in front of the middle of the used elements.

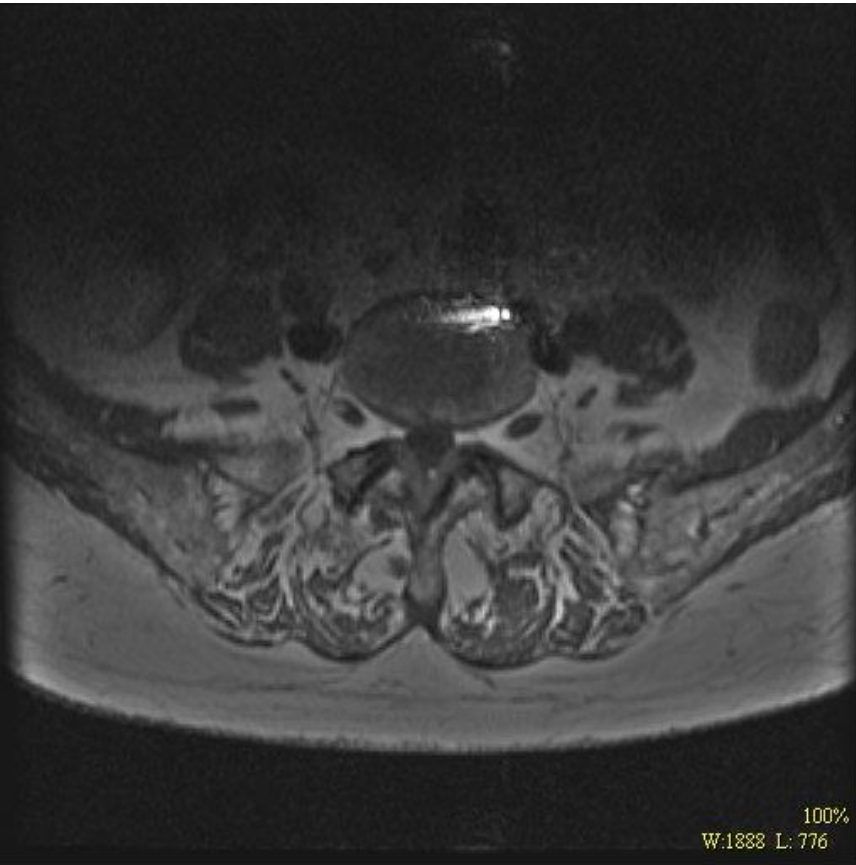
Each coil segment covers 12cm, so use:

FOV 24cm for 2 segments (example: CS12)

FOV 36cm for 3 segments (example LS456)

FOV > 42cm for 4 segments (example CTLMID)

Startifact



Issue description:

Peripheral signal artifact commonly called “startifact” occurred on axial plane.

Solution:

Customer needs to use the recommended methods of coil selection for field of view to minimize the possibility of seeing this artifact (see page 13).

Maxwell Term Artifacts



Issue description:

Artifact due to running flow compensation while phase and frequency are swapped. Flow compensation is applied on the frequency axis with phase and frequency swapped, flow comp. is applied A-P. Flow comp pulse generates a large Maxwell term down the Z axis (S-I) and causes ghosting, blurring and signal void towards the ends of the field of view. The bigger the field of view the worse the artifact.

Solution:

Simply turn off flow comp when phase and frequency are swapped. There is no benefit of running flow comp in this scenario since the compensation is no longer in the direction of flow.

Gibbs Artifacts

Issue description:

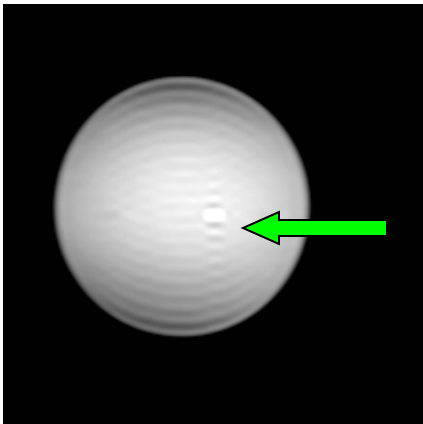
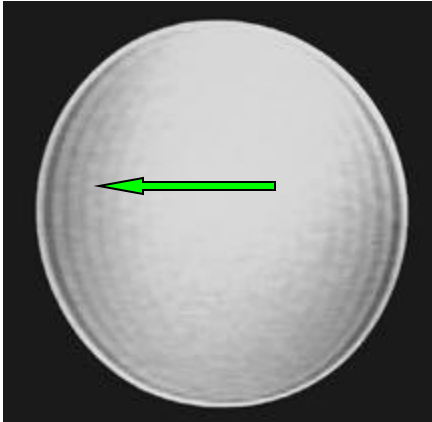
Gibbs ringing is a series of lines parallel to a sharp intensity edge in an image.

Root cause:

Ringing caused by incomplete digitization of the echo. Signal has not decayed to zero by the end of the acquisition window, and the echo is not fully digitized. Artifact in images when a small acquisition matrix is used. Therefore, the artifact is more pronounced in the 128 point dimension of a 512x128 acquisition matrix.

Solution:

Increase matrix size up to a squared one.



Fineline Artifacts

Issue description:

“Fineline artifact”. You may need to look close to see these since it is not real obvious. Here is some additional info regards to this artifact:

- See most on FSE-IR scan.
- Small FOV scan, thin slice.
- Very bony structure, such as foot, knee.
- Caused by FID of the 180° RF pulses.
- ZIP option can enhance the finelines.

Solution:

• **Elimination of Fineline Artifacts**

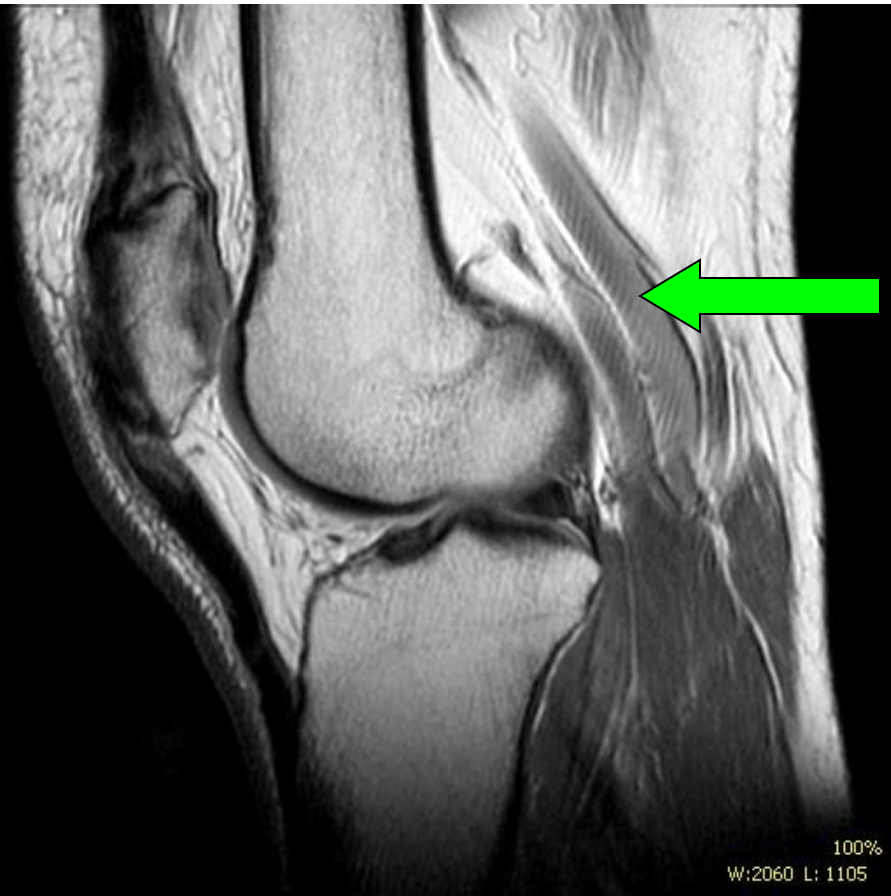
Use true even NEX, for example:

use 2 NEX;

Or use 4 NEX when NPW is selected. (Since NPW is selected, 2 NEX is actually a 1 NEX scan produces a fineline artifact.)



Fineline Artifact



The image above is a an example of “Fineline artifact”.

Twinspeed Zoom Artifacts

This is one example of a TwinSpeed ZOOM artifact. Looks like an RF zipper but is caused by peripheral signal excited outside of the FOV. Need to utilize applications workarounds in order to minimize this (see page 13).

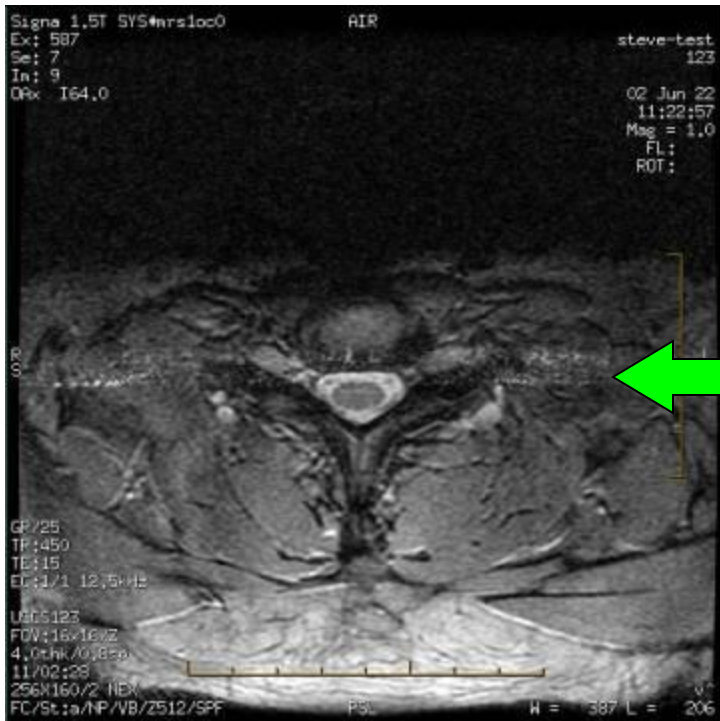
Some notes:

Spatial SAT excites tissue beyond the Zoom coil - The Crusher gradient from the Zoom becomes ineffective.

3-Coil CTL (LS456) in Zoom Mode can receive signal from tissue excited by the wide SAT pulse outside the FOV, and wrap into the phase direction.

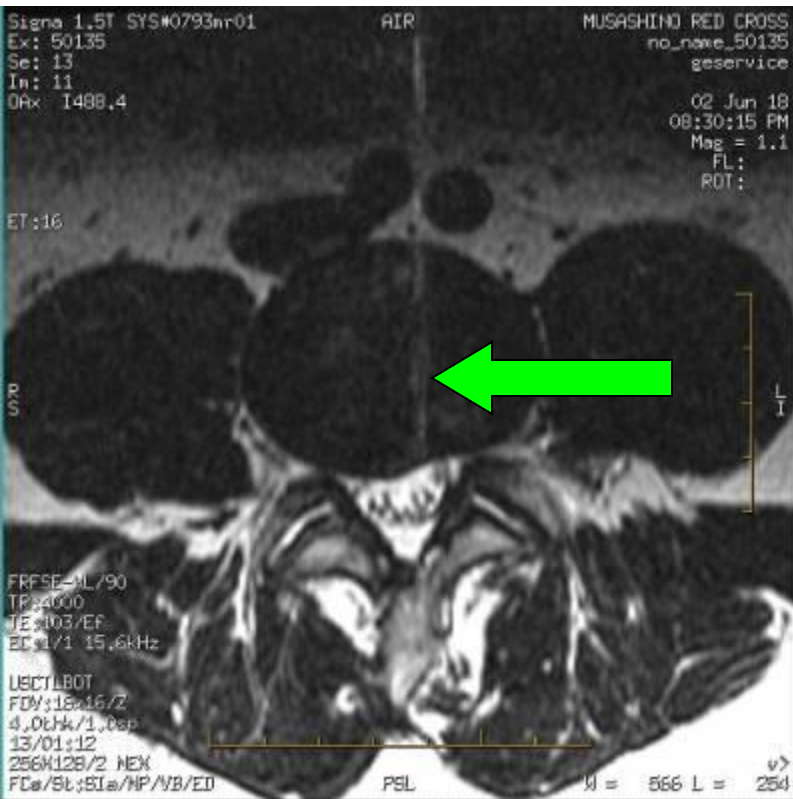


Twinspeed Zoom Artifacts/CSpine



An anterior spatial SAT excites the tissue beyond the zoom coil and the crusher gradient from the zoom coil becomes ineffective. 3-coil CTL (CS123) Zoom mode receives a signal from the tissue, excited by the SAT pulse outside the FOV, and wraps into the phase direction. Need to utilize applications workarounds in order to minimize this (see page 13).

Twinspeed Zoom Artifacts/LSpine



In this case, a spatial SAT excites the tissue beyond the zoom coil and the crusher gradient from the zoom becomes ineffective. 3-coil CTL (LS456) Zoom mode can receive signal from tissue, excited by the wide SAT pulse outside the FOV, and can wrap into the phase direction. Need to utilize applications workarounds in order to minimize this (see page 72).

Twinspeed Zoom Artifacts/LSpine

Application Tips for Reduction of Twinspeed Zoom Artifacts:

Zoom mode introduces artifacts by the introduction of signal received outside the **Zoom** gradient limit. This is less of an issue with the **Whole** gradient mode.

The artifacts can be reduced by using the following methods:

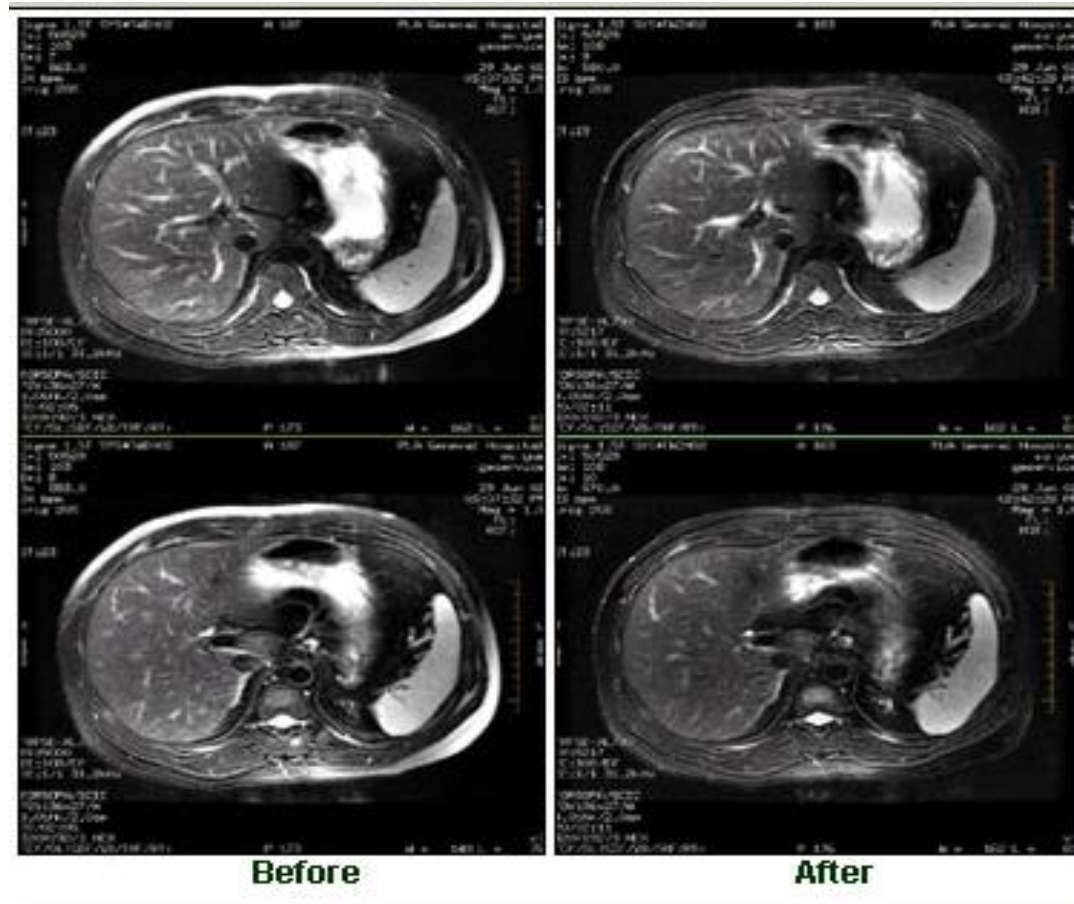
- Eliminating or narrowing the SAT pulse width (30mm)
- Selecting proper coils (inside the zoom limit, see page 66)
- In almost all cases, using Whole Mode will eliminate these artifacts.

Note: When the scan FOV allows, a 2-coil CTL has approximately 27% more SNR than the 3-coils

Twinspeed Fat Sat : Suggested Improvement

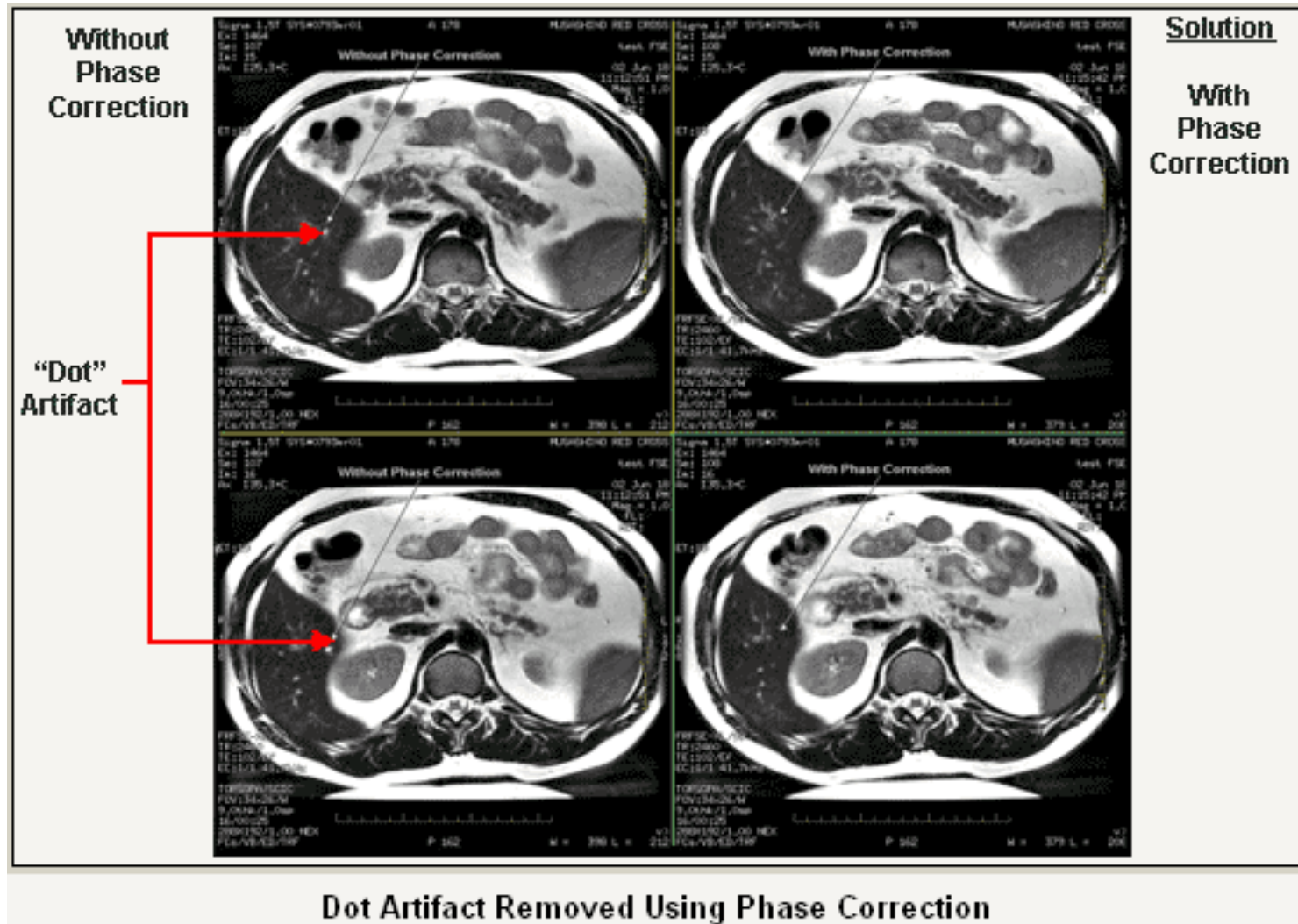
For a Fat Sat, use a high-order shim.

The graphic below displays the comparison between Fat Sat before and after applying a high-order shim.

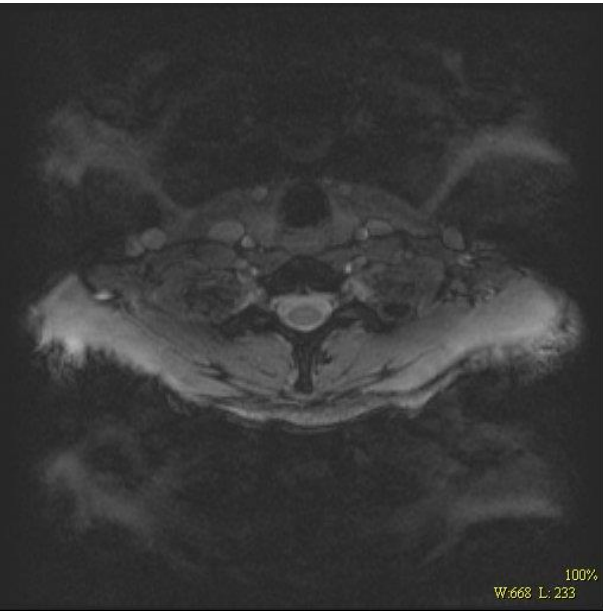


Specific IQ Issues: Dot Artifact

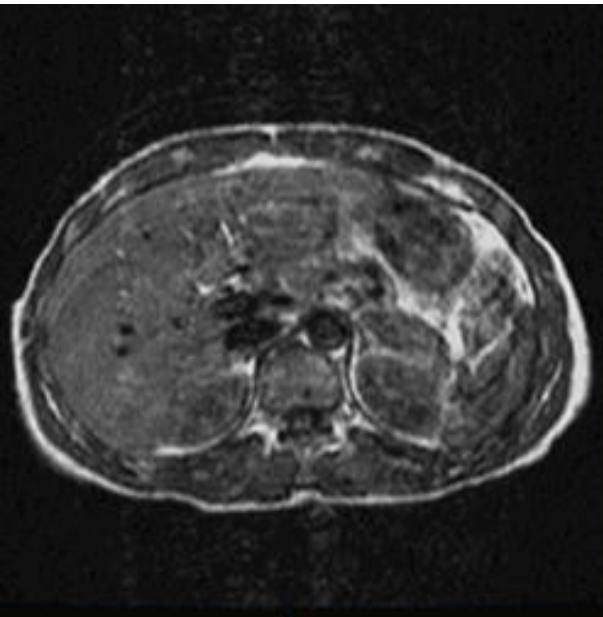
The graphic below displays the removal of dot artifact using phase correction.



Patient Respiratory Motion Artifacts



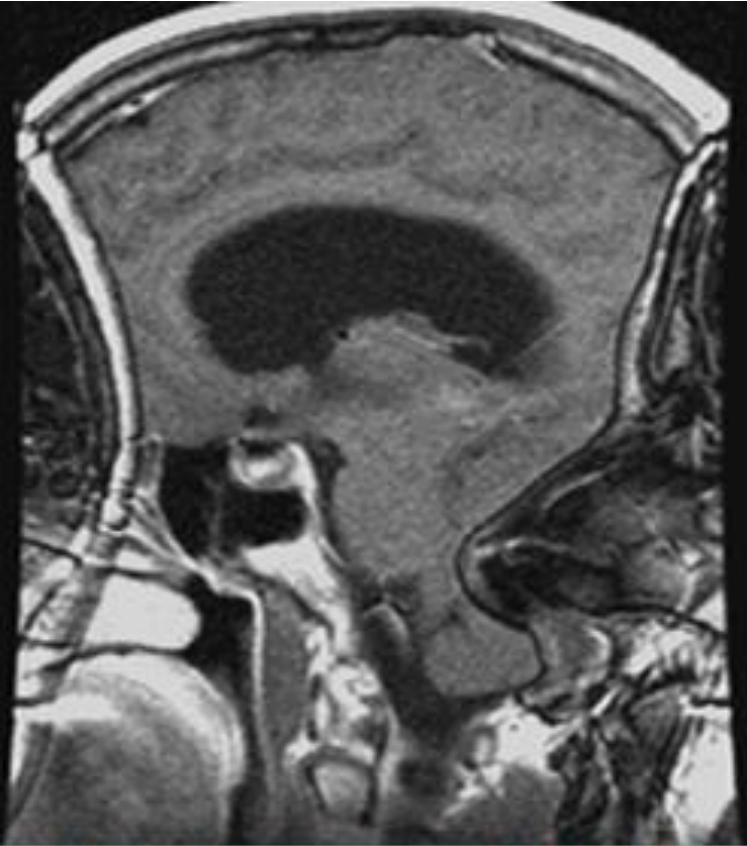
Ghosting seen in the spine image was isolated to 1 patient and is believed to be due to patient respiratory motion. Note the ghosts are well defined suggesting that the disturbance had a regular frequency.



Phase Wrap (Aliasing)

Issue description:

Anatomy outside the field of view "wraps" into the opposite side of the image.



Solution:

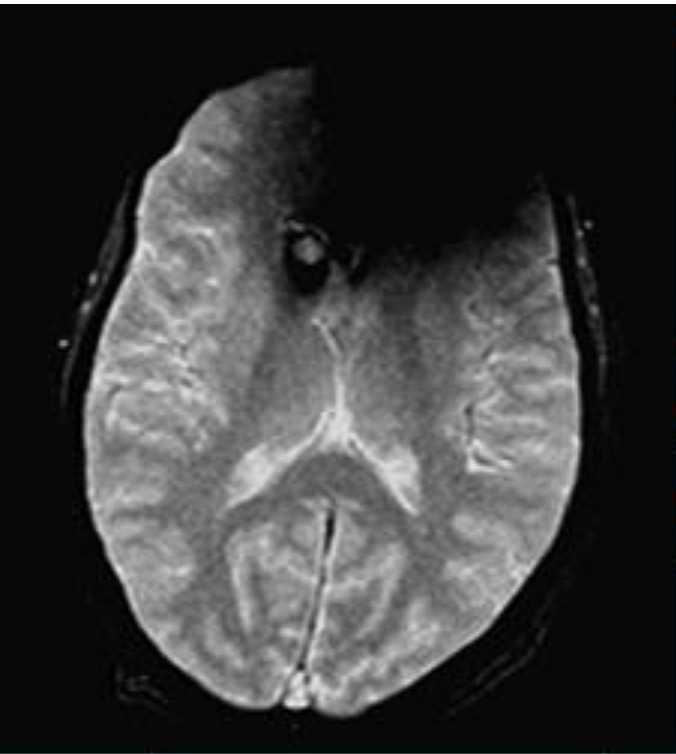
Corrected by scanning with a larger field of view, applying pre-saturation pulses to the undesired tissue, or by correctly phase encoding the extraneous tissue then discarding that information.

Note:

No Phase Wrap options typically do the latter by doubling the field of view, doubling the number of phase encodes (to keep resolution constant) and halving the number of averages (to keep scan time constant) then discarding the additional data and processing the image within the desired field of view.

Other Applications Related Artifacts

Gradient Echo Sequence Showing Magnetic Susceptibility



Issue description:

The large artifact seen anteriorly in this patient was caused by dental fillings.

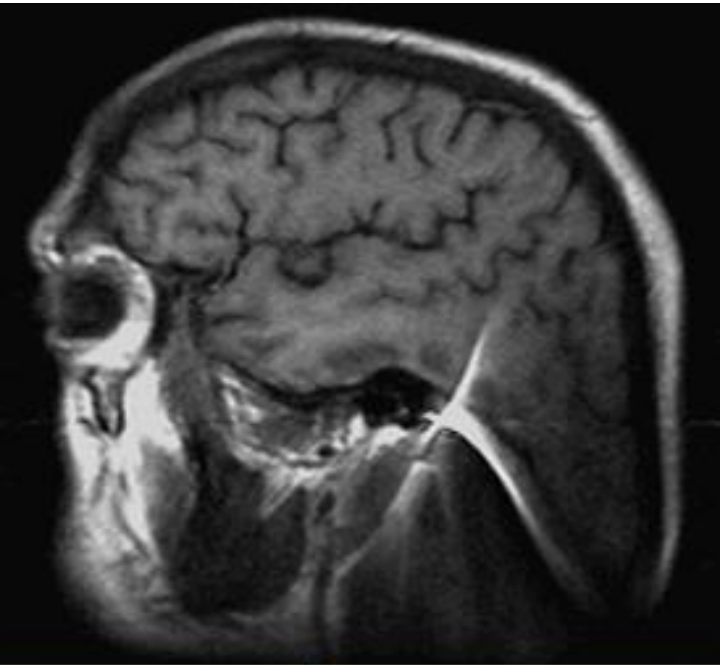
Root cause:

Gradient echo sequences (those using a gradient reversal to rephase the transverse magnetization instead of a 180° RF pulse) do not re-phase the decay caused by local field in-homogeneities. Metal causes such an in-homogeneity.

Metal Artifact Causing Cone-head Appearance

Issue description:

Metal Artifact Causing Cone-head Appearance.



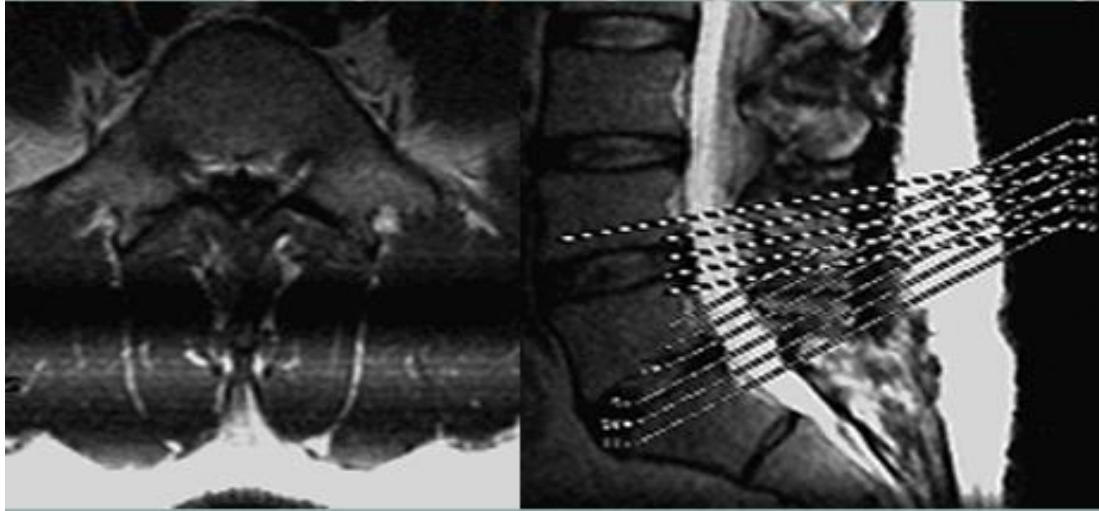
Root cause:

Evident distortion due to the presence of a small metal clip on an elastic in the patients hair. Ferrous metal will cause a magnetic field in-homogeneity which in turn causes a local signal void, often ringed by an area of high signal intensity, as well as a distortion of the image.

Solution:

Once the metal was removed the patients head returned to a more normal shape.

Axial/Oblique Images at Intersecting Planes



Issue description:

Axial/Oblique Images at Intersecting Planes.

Root cause:

Visible dark bands on the T1-weighted Spin Echo axial image are due to the intersection of other axial images through the image. These are acquired in a multi-planar fashion and thus cause pre-excitation (saturation) of the protons in the area where the slices intersect. Sagittal image illustrates where the slices were obtained and how they intersect posteriorly.

Metal Effects on Fat Saturation



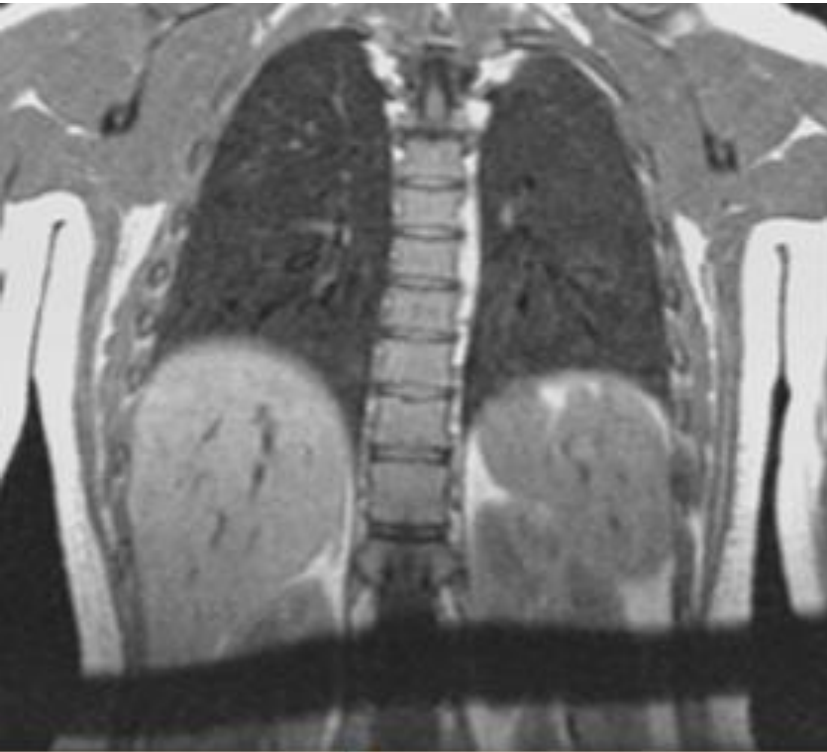
Issue description:

Metal Effects on Fat Saturation.

Root cause:

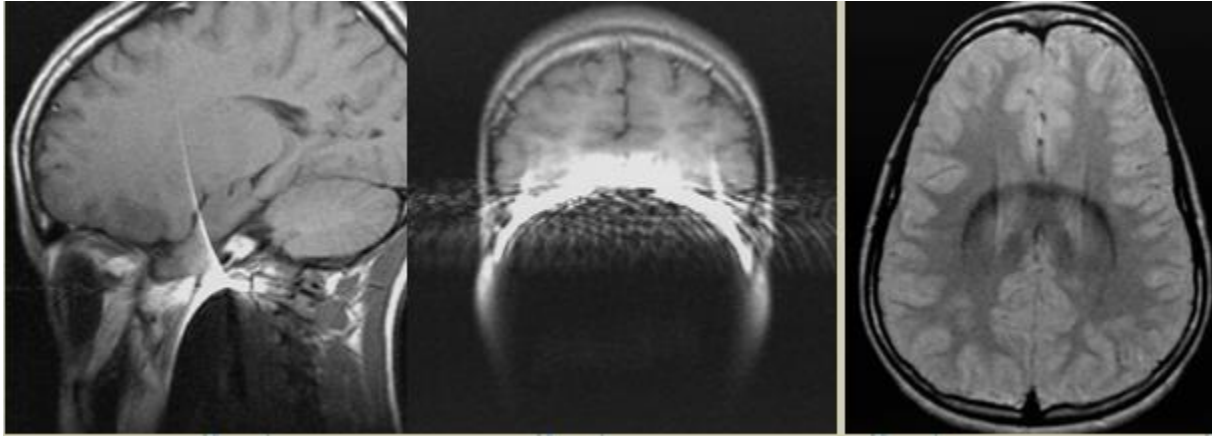
‘T2 weighted, Fast Spin Echo, Fat Saturated’ image shows how poor Fat Saturation can be in a region of Metallic Prosthesis. Caused by an alteration in the local magnetic field resulting in a change to the *precessional frequencies*, rendering the Chemical Saturation pulses ineffective.

Patient Wearing Belt



Patient was wearing a metal-studded belt during this coronal T1-weighted abdominal scan.

Braces



Braces may cause an artifact distant from the source. Axial proton density (left) and T2-weighted (right) images exhibit horseshoe shaped artifacts. Note it is much subtler on the T2 image. Another example below with breast imaging.

Pant Zip

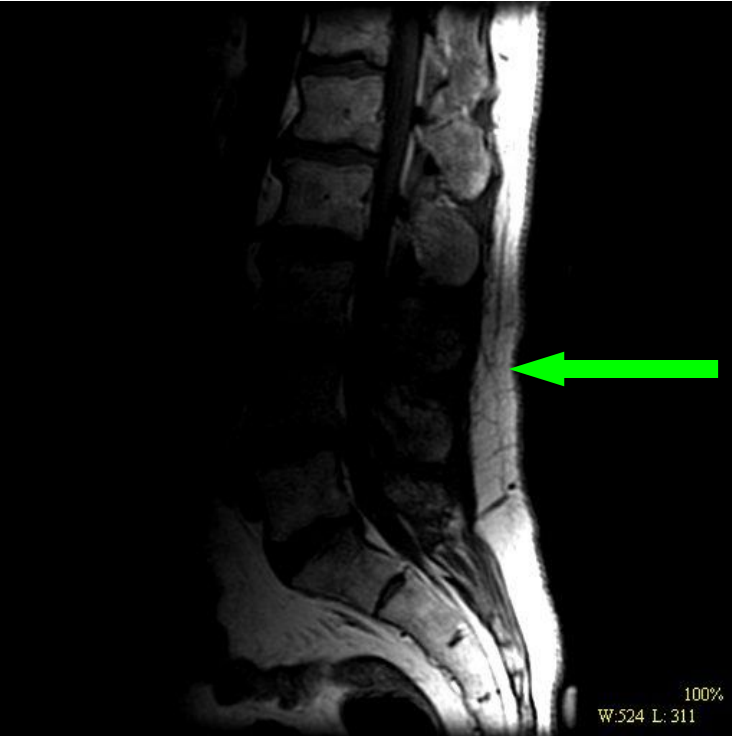


Pant zip may cause an artifact distant from the source. Here an example with breast imaging.

MR Artifacts Library

Shading/Signal Void
Coil Related Artifacts
RF Related Artifacts
Corduroy Artifacts
Ghosting/Stability Artifacts
MGD Related Artifacts
TPS Related Artifacts
Gradient Related Artifacts
Coherent Noise
Multiple MR System Cross-talk

Shading/Signal Void



Issue description:

The signal void seen here with the CTL coil looks like a possible dead channel.

Root cause:

No multi-coil errors

Must have been a poor connection to the pin that is connected to the channel with very low signal.

Solution:

Corrected by simply reseating the connector at the quick disconnect.

Shading/Signal Void

Issue description:

Signal void.

Root cause:

Operator error

C Spine series first with CS12 config and Inferior sat band on

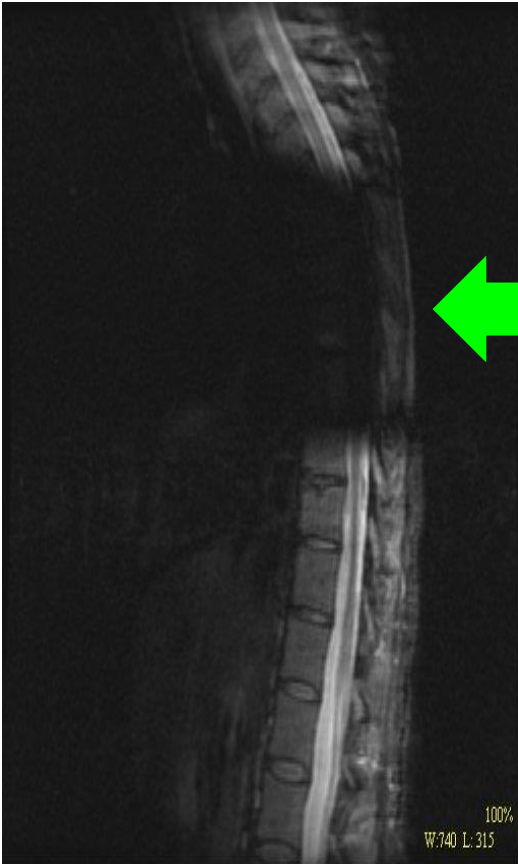
Then “RX Scan” with a large Superior table delta and CTLMID for Thoracic area.

The void seen here is the Inferior Sat Band placed on the earlier C Spine scan.

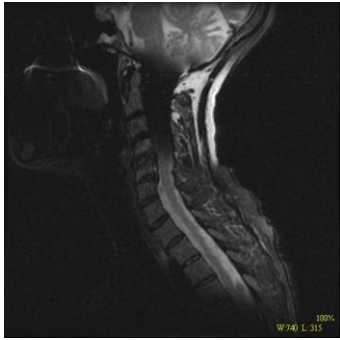
Solution:

“RX Scan” will keep the same location of the sat band in the FOV on T Spine scan.

Do a “RX Series” or perform new scan with a new landmark.

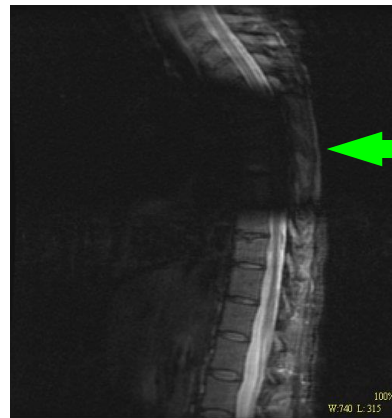


Shading/Signal Void



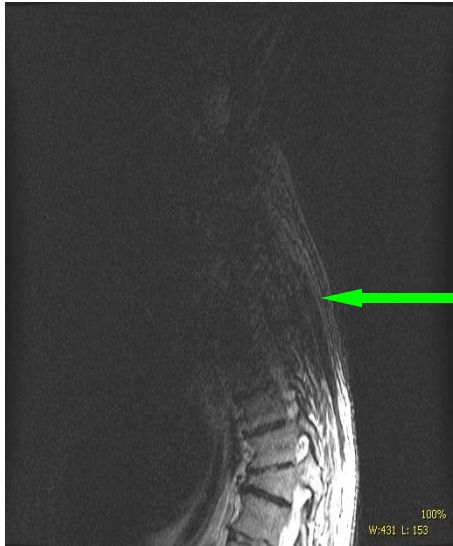
C Spine scan

T Spine scan with
inferior sat band
from the C Spine scan



T Spine scan
repeated without sat
band on.

Shading/Signal Void



Issue description:

Shading on C Spine scan and CTL coil..

Root cause:

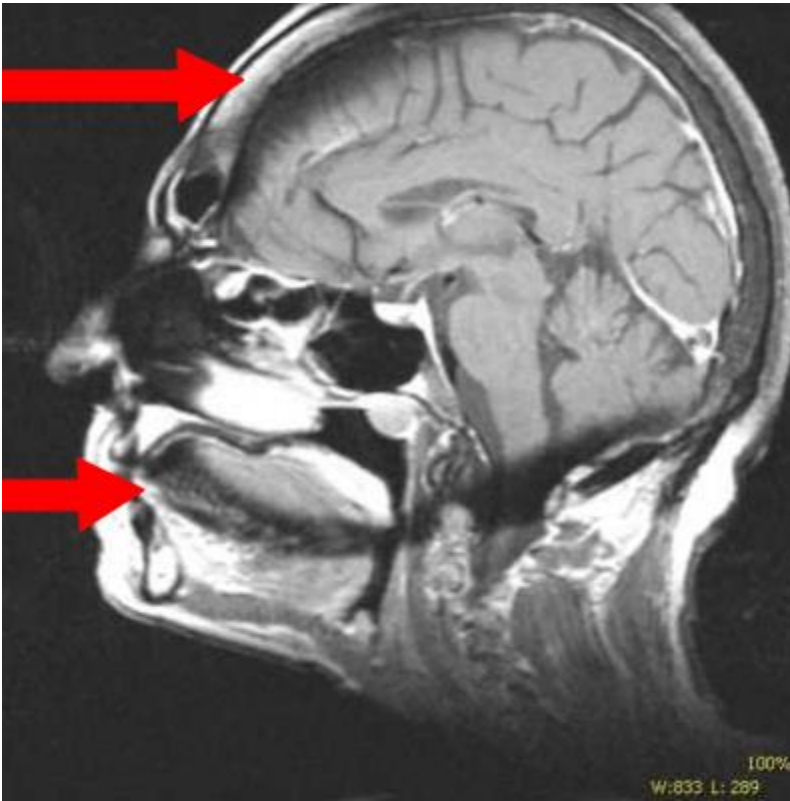
Landmark done for C Spine
T Spine protocol and CTLMID config on
So top 2 coils are turned off.

Solution:

Corrected with
CTLTOP config.



Shading



Issue description:

Shading.

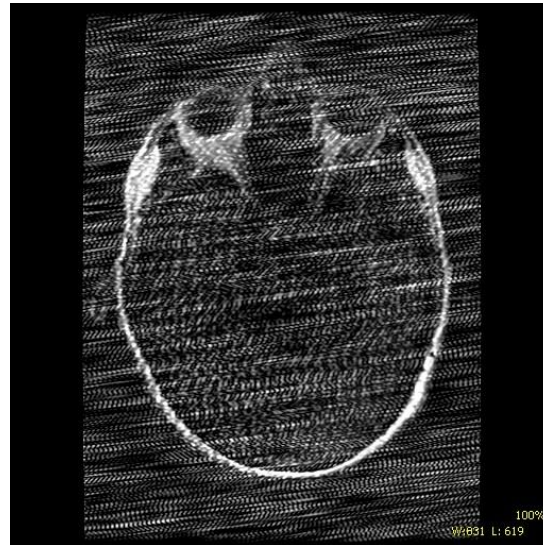
Root cause:

FAT SAT on. The center frequency off by about 100Hz, this caused the SAT pulse to saturate some of the water frequency.

Solution:

Run MPS (Manual Pre-Scan) and re-adjust water peak

Coil Related Artifacts



Issue description:

T2 Flair with 8 Ch NVARRAY coil.
Intermittent but frequent, once a day,
sometimes vary in appearance
Only seen with NVARRAY coil.

Root cause:

Bad 8 Ch NVARRAY coil.

Solution:

Coil replacement.

RF Related Artifacts

Issue description:

System with solid state RF amps.
Ghosting.

Root cause:

Bad teli RF amp. SPT stability failed.

Troubleshooting:

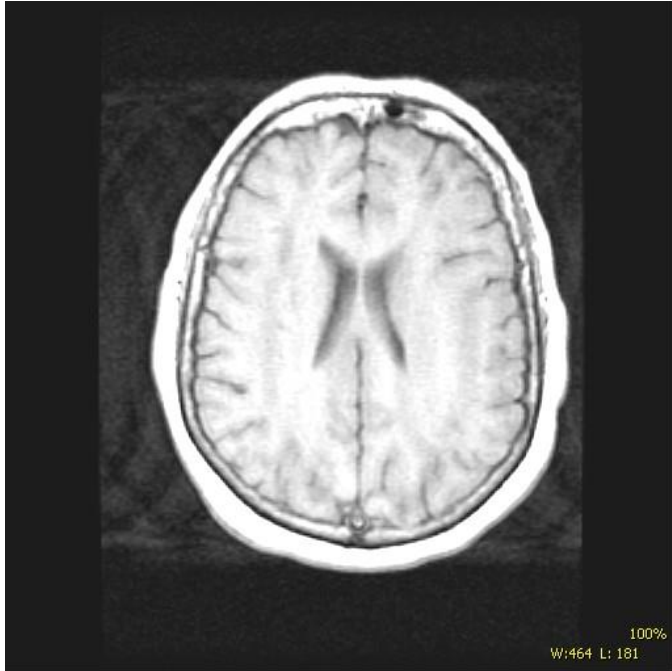
Able to isolate the bad amp by scanning with the HEAD coil with one amp at a time (one amp run into a dummy load)

Can do this with the HEAD coil,

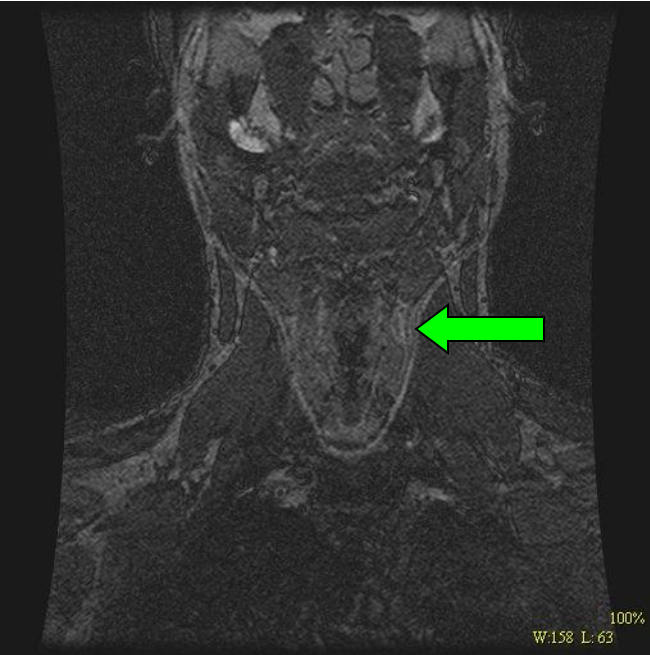
Can't do this with the BODY coil because the dump resistor in the RFI will overheat.

Solution:

Replace bad RF amp.



RF Related Artifacts



Issue description:

Coronal 3D MRA with NVARARRAY coil, and severe slice wrap: back of the head and neck wrapping into this image.

Root cause:

RF stability issue.

Failures with RFT which led to a couple of things.

Troubleshooting:

SPT passed.

EFB bypassed on the APM, the RFT failures cleaned up and the slice wrap disappeared.

APM board replaced but didn't completely resolve this.

Found that BODY Directional coupler was causing some of the RFT failures.

Solution:

After the replacement of the APM Board and the BODY Directional coupler, the problem was resolved.

RF Related Artifacts



Site has a TwinSpeed, image on the left, no slice wrap.
And an EchoSpeed, image on the right with slice wrap.
Same exact protocol used with 2 different patients.

Corduroy Artifacts



Issue description:

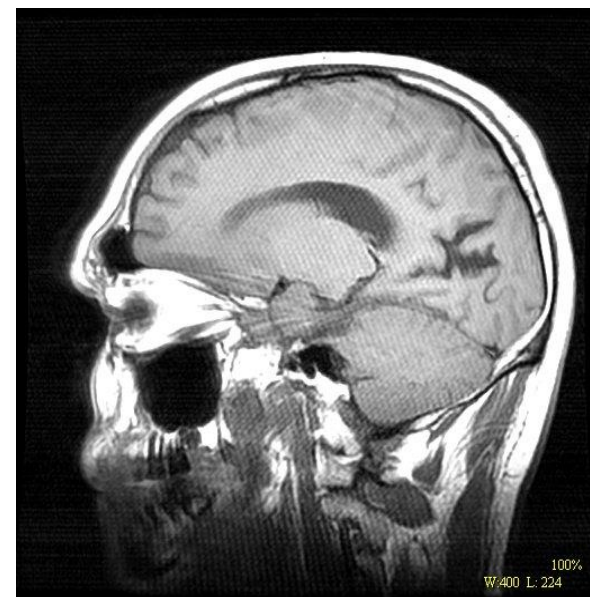
Typical Corduroy artifact.

Troubleshoot:

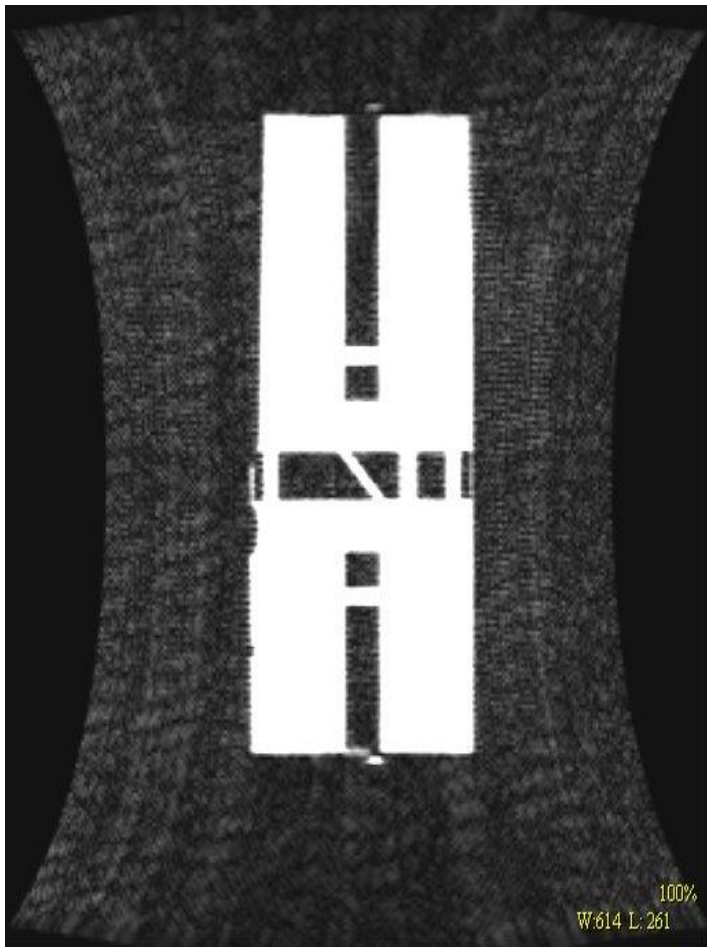
EPI white pixel test.

Solution:

Find and eliminate the corduroy artifacts source.



Corduroy Artifacts



Issue description:

Cardiac system with an LCC magnet.
Corduroy artifacts.

Root cause:

Spike noise issue caused by loose gradient coil bolts at the rear of the CRM coil.

Solution:

Tighten gradient cable connection.

Ghosting/Stability Artifacts

Issue description:

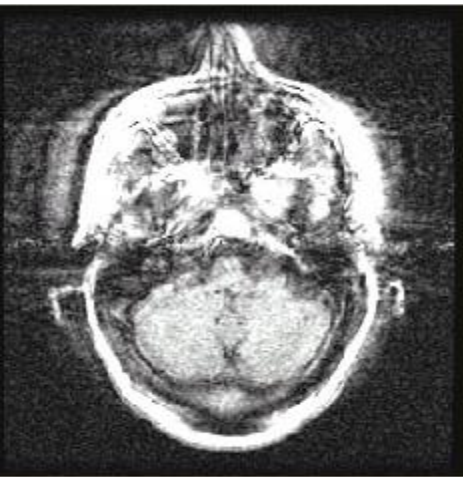
Ghosting artifacts.

Root cause:

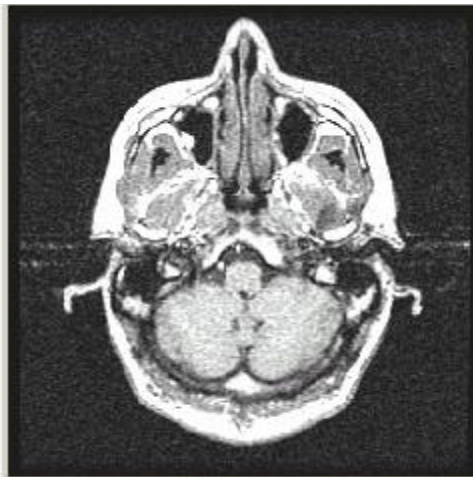
Patient moved during the procedure (example 1). Notice that the ghosting is evident in the nose and face area but not towards the back of the head.

Solution:

Example 2 is a repeat of the same scan, but this time the patient did not move.



Example 1



Example 2

Ghosting/Stability Artifacts

Issue description:

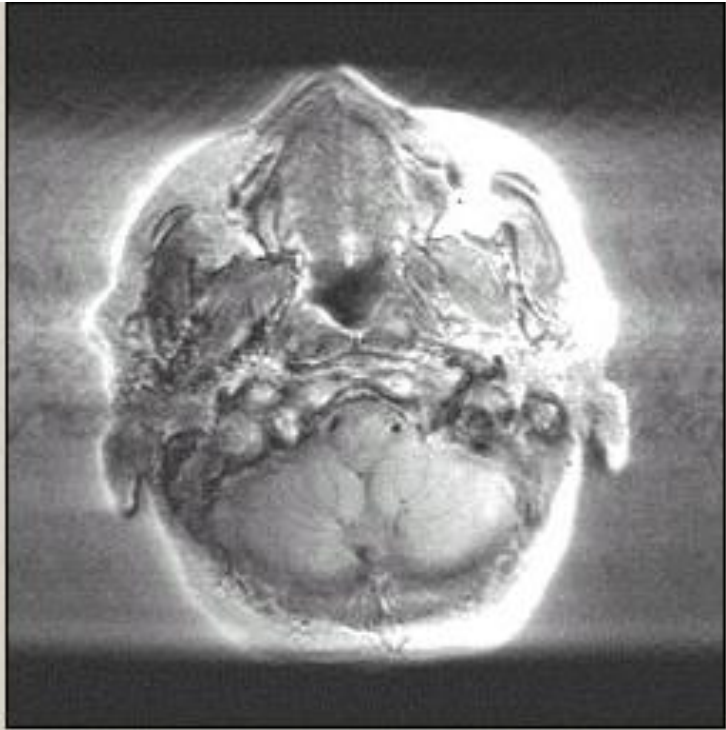
Ghosting artifacts.

Root cause:

Ghosting in this image is not due to motion. It is from an actual system failure. Notice that unlike the ghosts due to motion (examples 1 and 2, page 14), a ghost from a system problem will not be restricted to the anatomy in motion. It is across the entire piece of anatomy.

Solution:

Go thru the Ghosting Stability Flowchart (SPT, Gradient Stability, RF Stability, and Vibration Test)



Ghosting Artifacts Due to System Failure

Ghosting/Stability Artifacts

Issue description:

Ghosting.

Root cause:

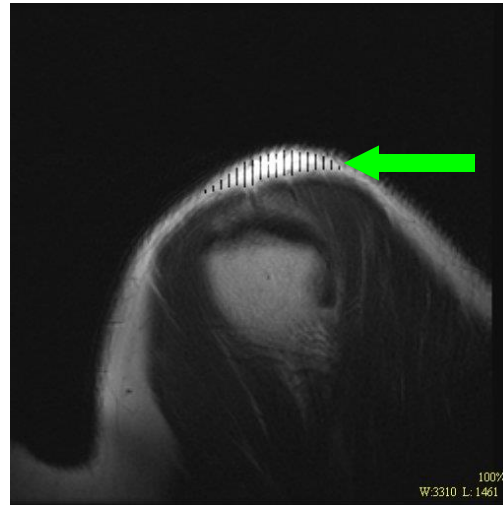
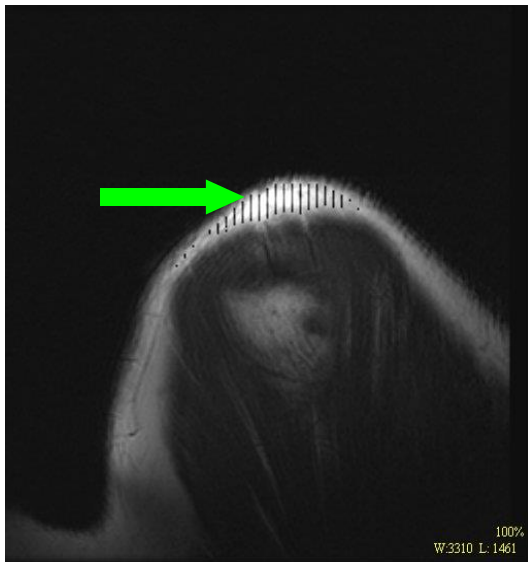
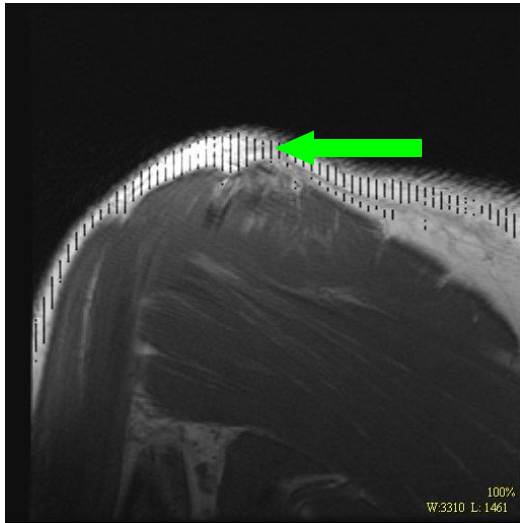
Ghosting artifacts due to respiratory motion in head images (often seen on sedated patients, who have a deep, regular cycle in their breathing pattern). This is sometimes harder to detect as a motion problem. In this example, note how the ghosts are well defined, suggesting that the disturbance has a regular frequency (cyclical).



Ghosting Artifacts Due to Respiratory

The source of cyclical disturbance can be anything that has a regular, repeating rhythm (cold-head, fans, air handlers, rotating machinery, and in some cases people with deep regular breathing).

MGD Related Artifacts



Issue description:

1.5T ES Excite and a Reflex 400 AP.

The artifacts with Shoulder PA coil and CTL coil.

Very reproducible, not at all intermittent. Only seen in areas with high signal, like fat.

Root cause:

Digital and possibly recon related.

Troubleshooting:

Slave AP board replaced in the MGD chassis for a different problem 2 days before. AP board very suspect based on this.

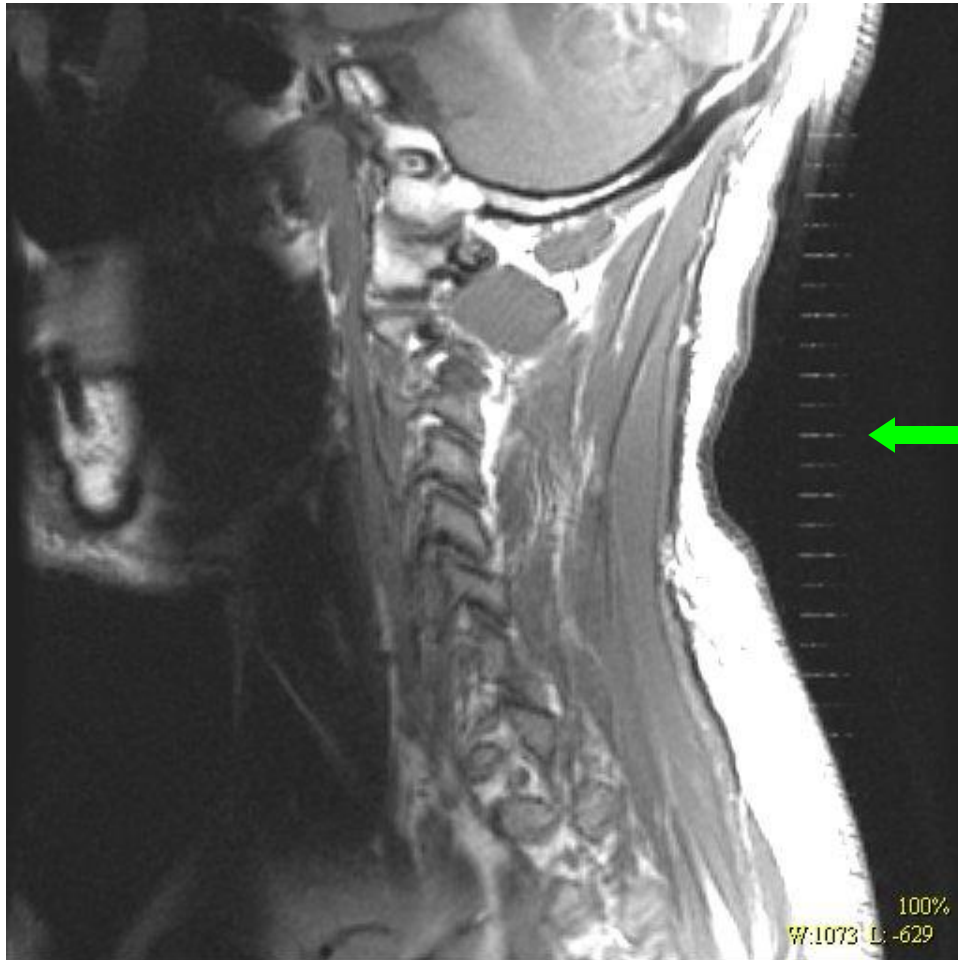
Slave AP board removed and run with only the Master AP in the Reflex 200 config. : no more artifacts.

Solution:

Slave AP board replaced.

One additional note, before we removed the Slave AP board we ran MGD Board level diags multiple runs and all passed.

TPS Related Artifacts



Issue description :

Recon Artifact.

Root cause:

Medcam Mother board.

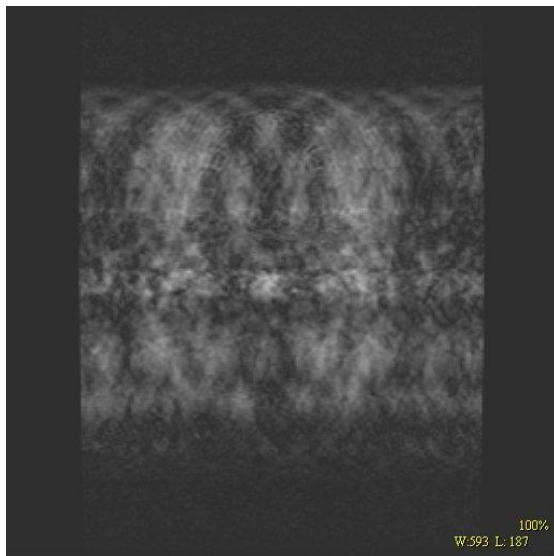
Troubleshooting:

TPS diags failed with Medcam errors.

Solution:

Replace the Medcam Mother Board.

TPS Related Artifacts



Issue description:

Intermittent phase smearing on images. Not really related to any particular type of sequence. The artifacts then started occurring on just about every scan.

Root cause:

CERD Digital Board

Troubleshooting:

SPT passed all tests.

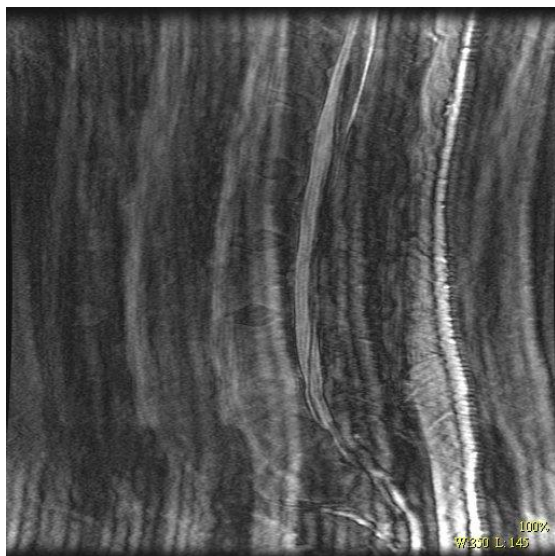
Run TPS diags.

CERD diags failed

All of the CERD errors pointed to functions on the CERD Digital Board.

Solution:

Replaced the CERD Digital Board.



TPS Related Artifacts

Issue description:

ES with SIII magnet.

FSE scan in the HEAD coil.

Very reproducible to varying degrees of ghosting, only on Head FSE scans.

Same sequence with the BODY coil and CTL coil: no problem. Other PSD's in the HEAD coil are not a problem.

Root cause:

Bad IPG board.

Troubleshooting:

SPT fail FSE stability with HEAD only

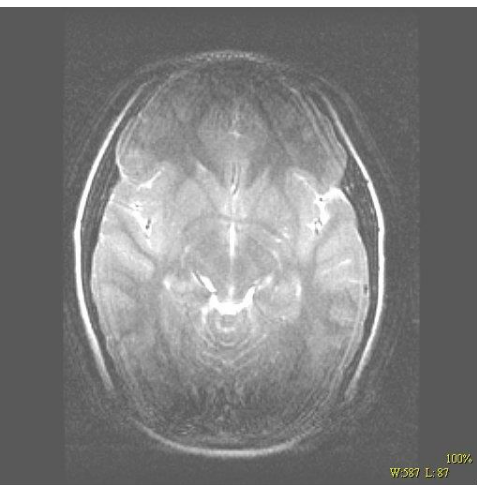
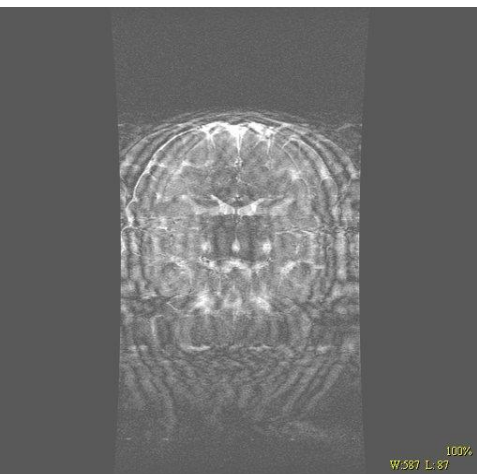
BODY FSE Stability passed.

FGRE Stability passed on both Head and BODY.

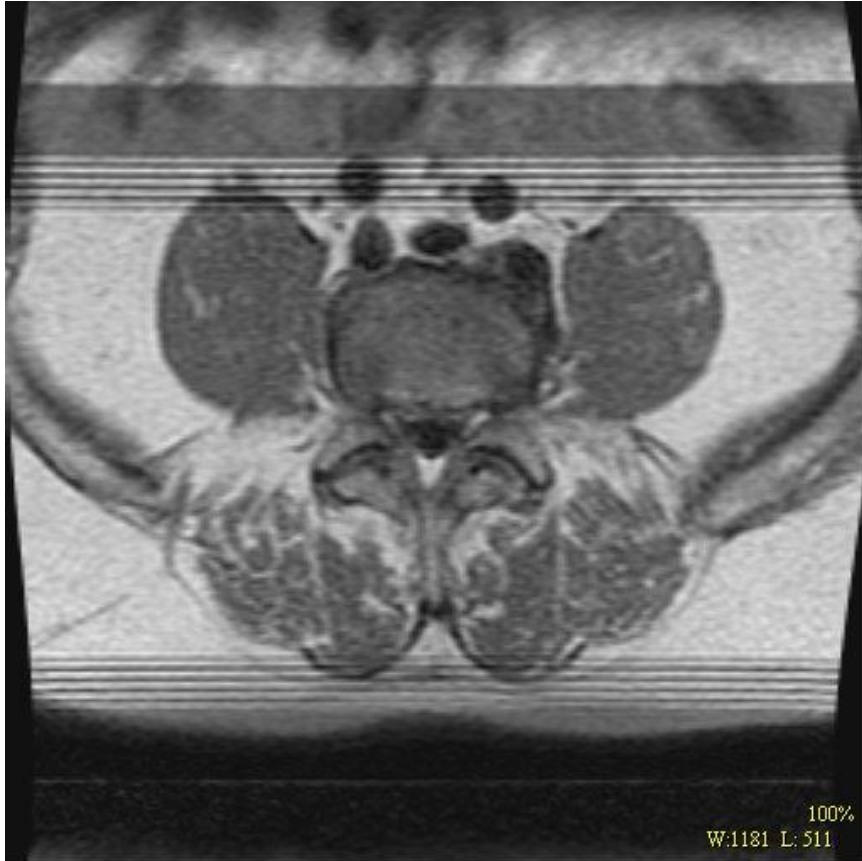
IPG had been replaced for a different issue at about the time the Customer started having these problems.

Solution:

Replace the IPG Board.



TPS Related Artifacts



Issue description:

Acquisition Artifacts.

Root cause:

CERD Digital board.

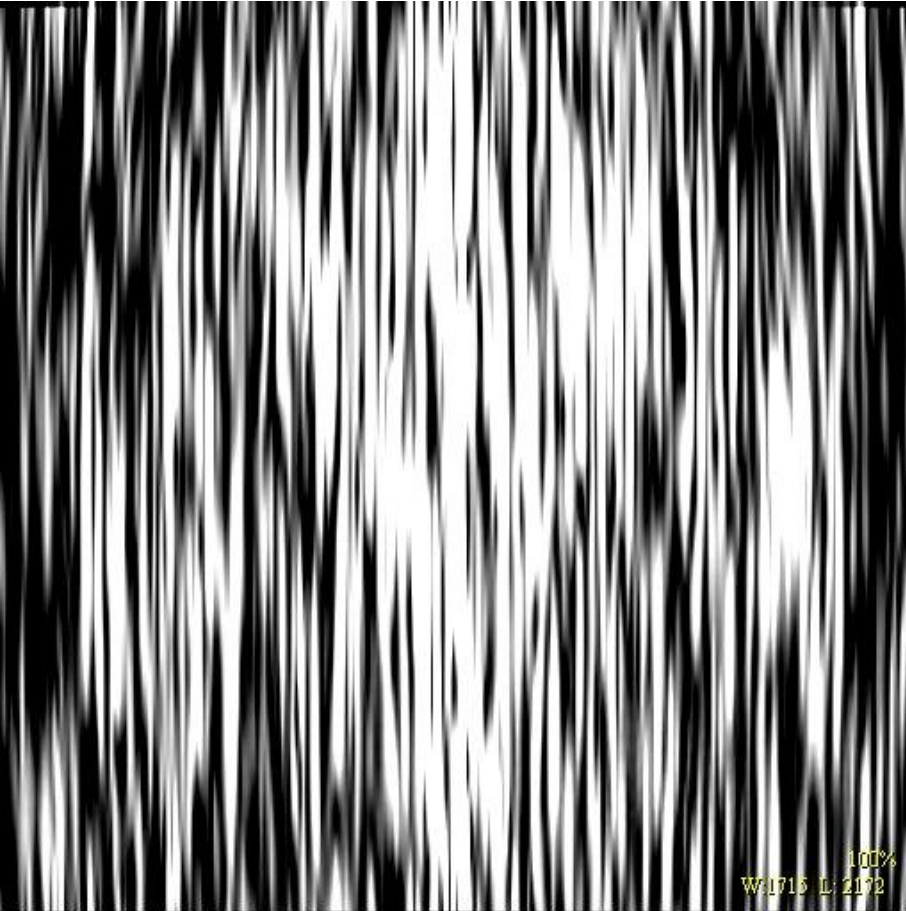
Troubleshooting:

No errors in the log at the time and all TPS diags pass.

Solution:

Replace the CERD Digital Board.

TPS Relates Artifacts



Issue description:

Artifact when using Multi-coil.

Root cause:

Bad receiver board in the TPS.

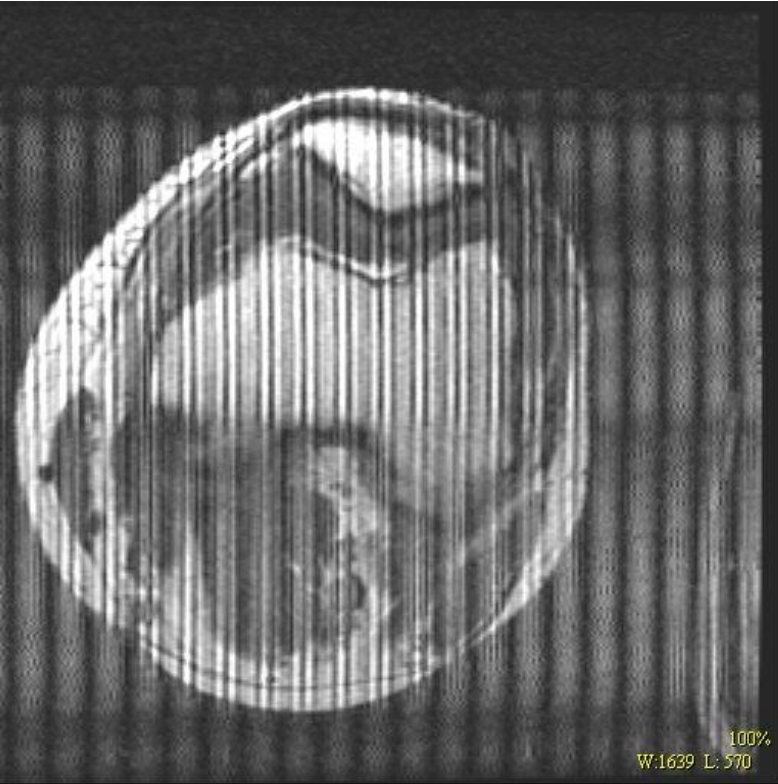
Troubleshooting:

Artifact only seen on 1 channel by running a scan using the CTL coil with CV “saveinter” set to 1. The only image that was bad in the intermediate images was with receiver 1 (0,1,2,3 scheme).

Solution:

Replace the bad receiver.

Gradient Related Artifacts



Issue description:

1.5T HS SGD Gradients.

Artifacts pretty repeatable.

Appeared to only occur on the Knee scans and some scans done with the TORSO coil.

Root cause:

Artifact caused by the X axis PPBM.

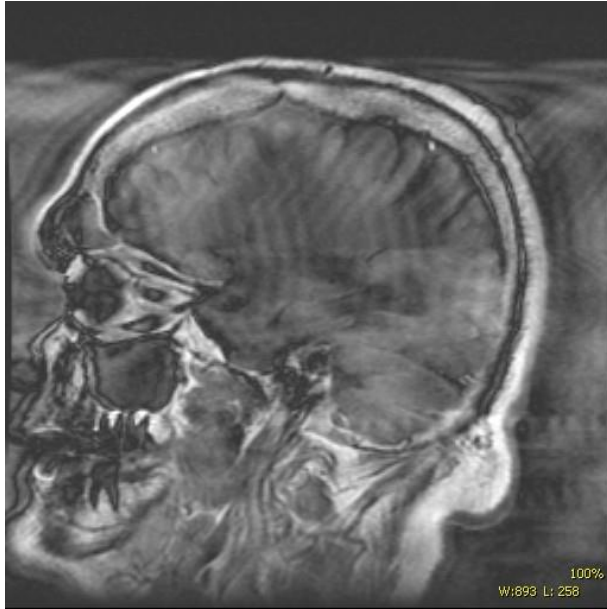
Troubleshooting:

Intermittent errors in the log stating X axis overload and PPBM errors.

Solution:

Replace the PPBM.

Gradient Related Artifacts



Issue description:

1.5T LX SGD Gradients.

The artifact seen below while scanning with the BODY coil was also occurring at this site due to the spike noise the arcing was creating.

Root cause:

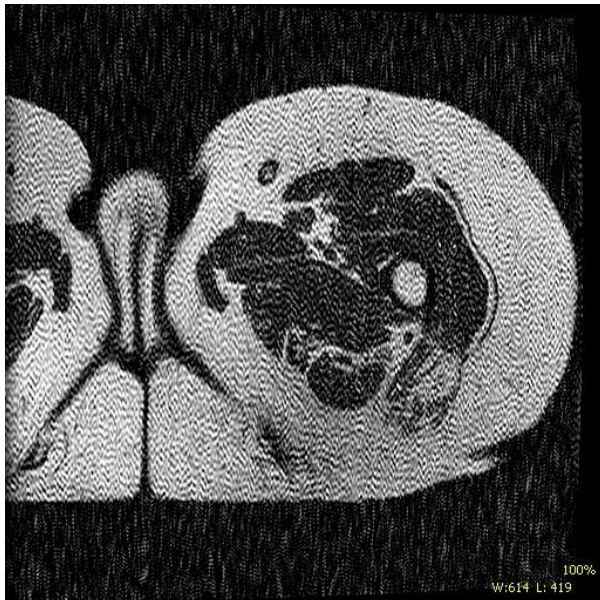
Ghosting seen in the image above was caused by a loose X axis Gradient drive cable on the magnet room side of the pen panel.

Troubleshooting:

This problem was also causing arcing at this loose connection.. One thing to note, the SPT failures were only on HEAD, BODY stability was completely clean.

Solution:

Tighten the gradient cable.



Gradient Related Artifacts



Issue description:

1.5T LX HS SGD Gradients.

C Spine scan with ghosting, but the customer was experiencing ghosting on pretty much everything.

Troubleshooting:

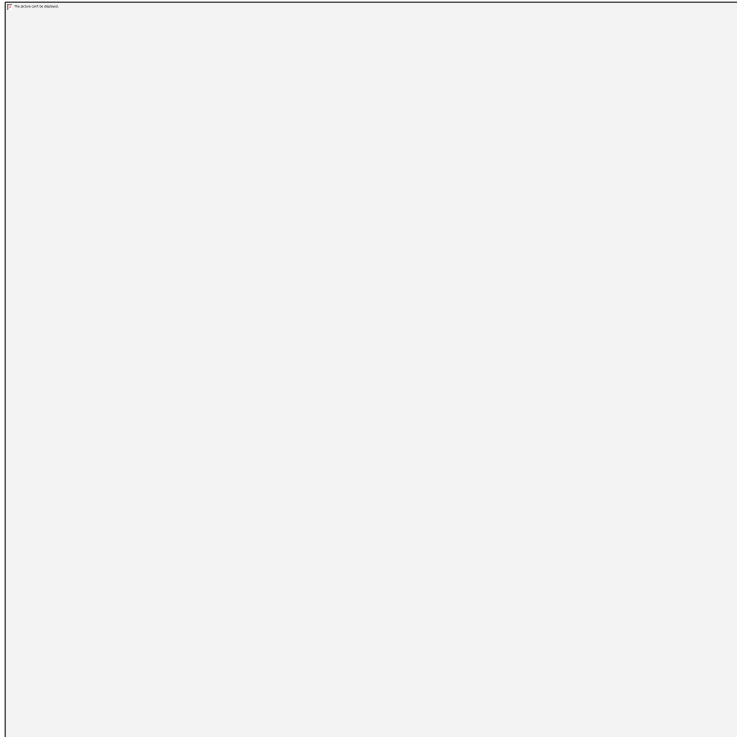
SPT showed some severe stability failures which were not as clear to interpret as most problems.

Caused by the Z axis minigram.

Solution:

Replace the Z axis minigram.

Gradient Related Artifacts



Issue description:

Artifact only on axial FLAIR sequence. Very repeatable (ran the same sequence in Sagittal and Coronal views and the artifact did not occur).

Troubleshooting:

Intermittent errors in the log stating a Y axis Master gradient overload.

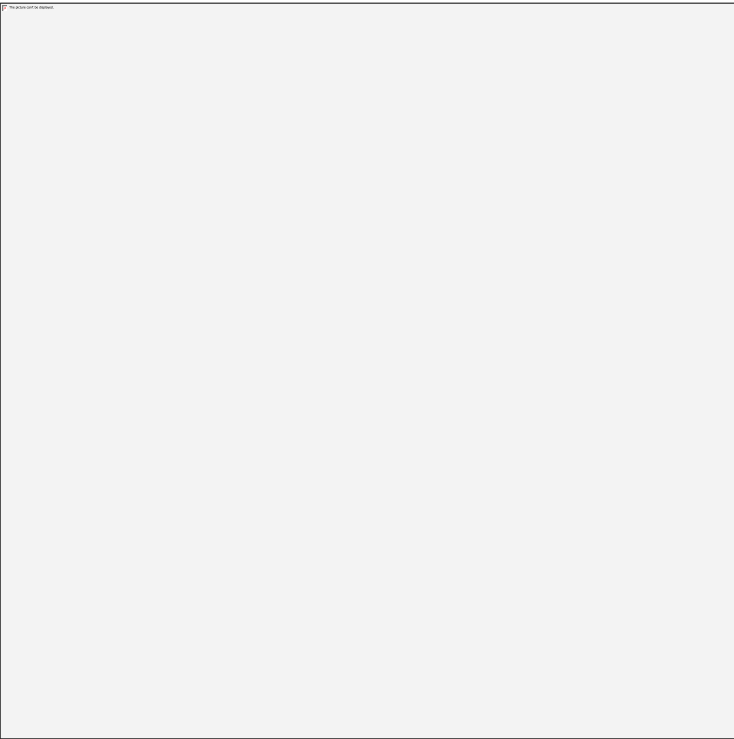
Swapping of the Y Master and Slave with other axis one at a time, problem isolated to the Y Slave gradient amp.

Y Master was evidently overloading because the Slave had marginal current drive.

Solution:

Replace Y Slave Gradient Amp.

Gradient Related Artifacts



Issue description:

1.5T ES SGD Gradients.

The Customer was able to press “start scan” to restart the acquisition. The artifact appeared.

Root cause:

Result of gradient errors occurring during the scan causing the scan to pause.

Troubleshooting:

Minigram and Minigram Power Supply suspected.

Gradient Related Artifacts

Issue description:

TW Excite 11.X.
Ghosting artifacts.

Root cause:

Due to a marginal SGA phase stability on the X axis.

Troubleshooting:

SPT passed but we could see an increased phase error when X was slice and increased echo shift error when X was readout.

SGA checked to be sure that the FMI's had been done, in particular the Gate Driver Board fix, and it was up to date on all modifications.

Solution:

X axis SGA replaced and the IQ issue was resolved.

Coherent Noise

Issue description:

Excite.

Artifacts on images when using the Invivo monitor.

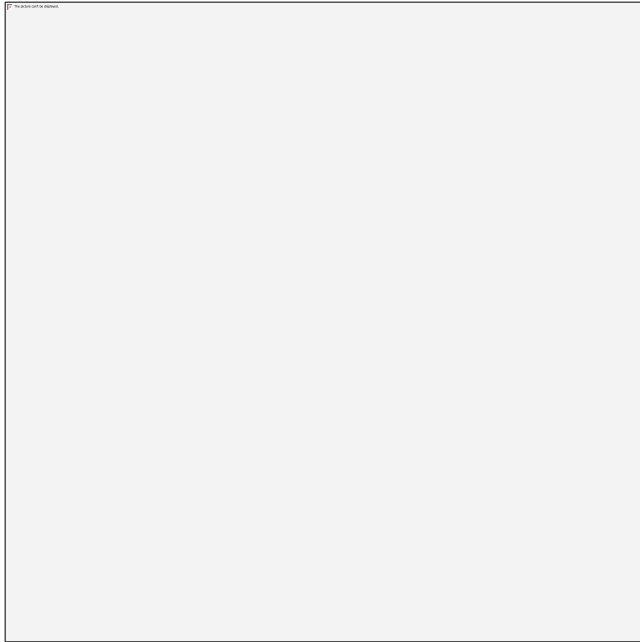
This site also has a 1.5T LX 9.1 SW that does not get this artifact when they use the same Invivo monitor.

Root cause:

Excite system is more sensitive to this noise from the Invivo system.

Solution:

Found out from Invivo that there are additional filters that may need to be added on the Invivo monitor when it is connected to an Excite system. The problem is with Invivo, not a problem with our system.



Coherent Noise

Issue description:

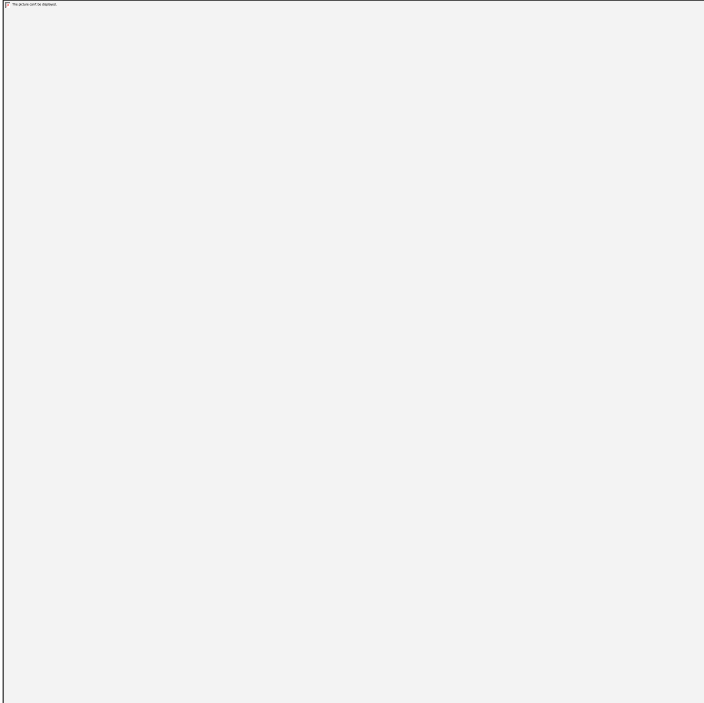
Multi MR system crosstalk from another MR at this site.

Root cause:

RF leak at the base of the screen room door. The Customer had waxed the floor earlier in the day in the operator room and had also waxed the copper base of the screen room door.

Solution:

The FE cleaned the wax off of the base of the Magnet room door and it cleared up this artifact. This RF leak allowed crosstalk into the magnet room from the other MR system.



Coherent Noise

Issue description:

1.5T LX with the Broadband spectroscopy option.
Zipper artifacts.



Root cause:

Zipper caused by the broadband spectro RF amplifier. The FE turned off the Broadband amp and the zipper went away.

Solution:

I am not sure what the final resolution was on this, but the point to make here is don't forget about the broadband amp if you have one and are seeing zippers

Coherent Noise

Issue description:

1.5T LX HS.

Zippers artifacts

Root cause:

Artifacts caused by a patient monitoring device that the Customer had.

This device was used for a long time at this site and had not been a problem.

Troubleshooting:

SPT coherent noise was failing very badly.

See results below.

Started by eliminating possible sources of RF in the room and after turning off this patient monitoring device, everything cleaned up.

Solution:

Device needs to be serviced in order to correct this problem.

Coherent Noise

1.5T COHERENT NOISE 256x256 R1 = 11

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mean	std dev	dot limit	zipper limit	mean zipper limit
------	---------	-----------	--------------	-------------------

84.294	43.414	431.609	344.780	87.294
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test	max	dot	max	zipper	max	mean zipper
------	-----	-----	-----	--------	-----	-------------

frequency	dot	frequency	zipper	frequency	mean zipper	frequency
-----------	-----	-----------	--------	-----------	-------------	-----------

63764000	1669.000	63774376	1669.000	63774376	730.167	63760580
----------	----------	----------	----------	----------	---------	----------

63789000	5638.000	63800356	5638.000	63800356	1777.171	63800356
----------	----------	----------	----------	----------	----------	----------

63814000	4645.000	63800320	4645.000	63800320	1869.081	63800320
----------	----------	----------	----------	----------	----------	----------

63839000	1022.000	63829472	1022.000	63829472	598.455	63829472
----------	----------	----------	----------	----------	---------	----------

63864000	1316.000	63875232	1316.000	63875232	756.273	63875232
----------	----------	----------	----------	----------	---------	----------

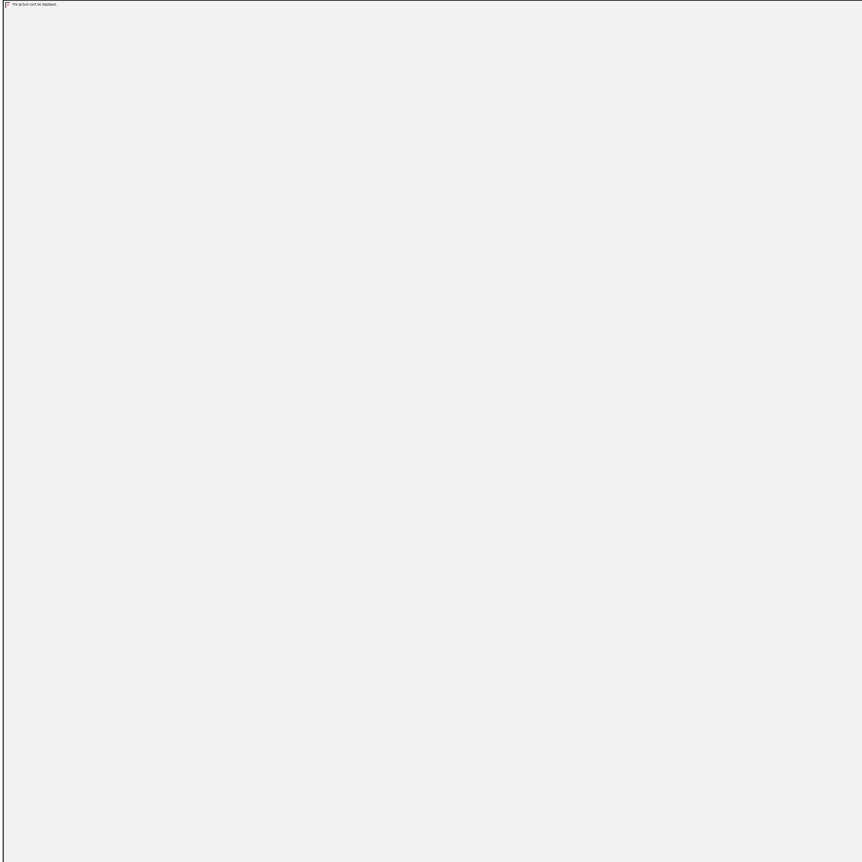
63889000	1604.000	63875320	1604.000	63875320	748.391	63875320
----------	----------	----------	----------	----------	---------	----------

63914000	1121.000	63912044	1121.000	63912044	656.211	63912044
----------	----------	----------	----------	----------	---------	----------

63939000	1017.000	63925812	1017.000	63925812	540.741	63925812
----------	----------	----------	----------	----------	---------	----------

63964000	770.000	63953376	736.000	63953376	495.083	63953376
----------	---------	----------	---------	----------	---------	----------

Coherent Noise



Issue description:

1.5T LX ES.

Coherent noise artifact

Root cause:

Crosstalk from another 1.5T MR system at this site.

Troubleshooting:

Check RF shielded room integrity.

Solution:

Fix room RF shield.

Multiple MR System Crosstalk



Issue description:

Multiple MR system crosstalk.

Solution:

The best strategy for eliminating this problem is separation of system frequencies, and good screen room integrity.

EPI Artifacts

EPI Ghosting Artifacts

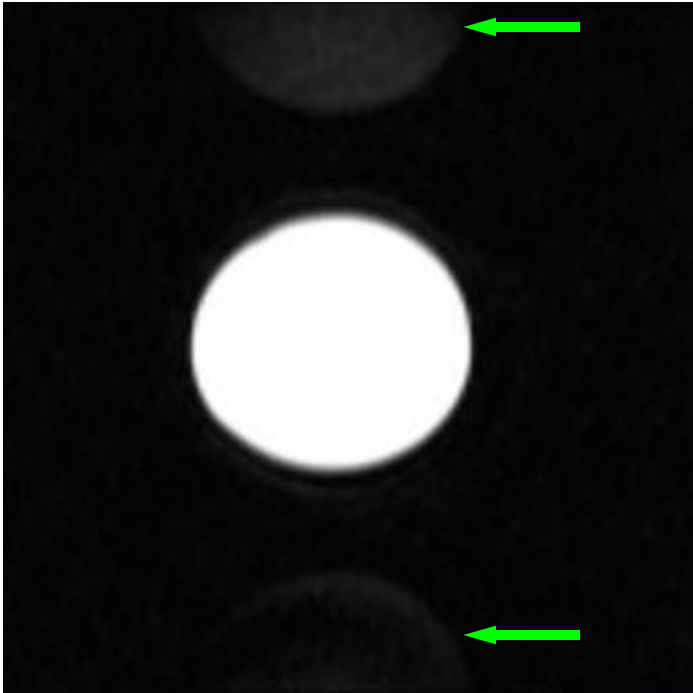
EPI and DWEPI Image Distortion

EPI Wormhole Artifacts and Signal Void

Slice To Slice Intensity Variation and Curved Sat Band

Other EPI Artifacts

EPI Ghosting Artifacts



Issue description:

1.5T LX ES.

Artifact example is an image from the B0 image test in EPT. At the install of this system, the B0 image failed.

Root cause:

A shorted RF shield in the BRM caused this.

Solution:

Replace the BRM.

EPI Ghosting Artifacts

Issue description:

Twinspeed system.

DWEPI combined image in the ZOOM mode.

Ghost seen around the edges of this image.

Root cause:

Due to a crossterm eddy current.

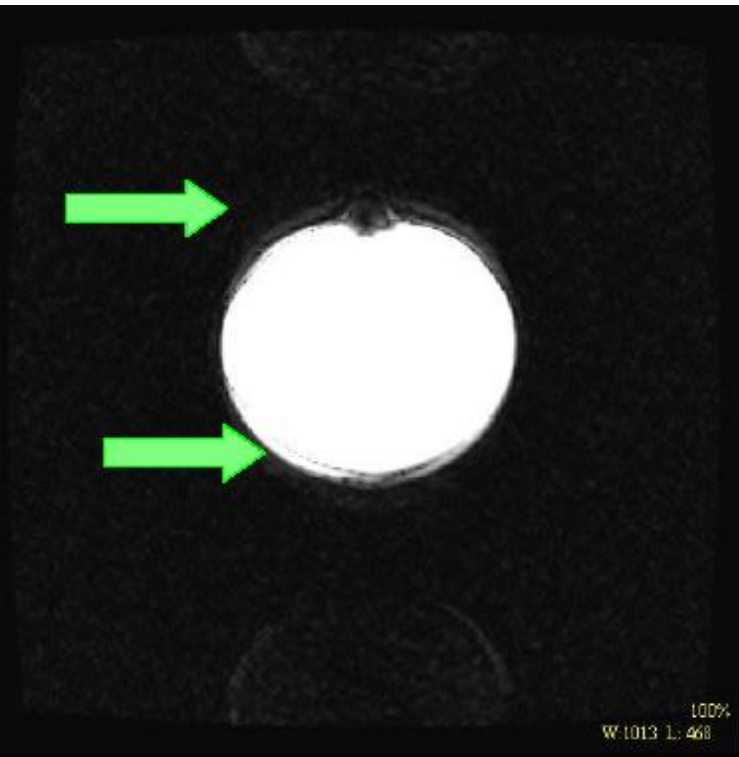
Troubleshooting:

Grafidy done numerous times in attempts to resolve this and it came into spec every time. This did not have any impact on the images though. Ran the DWEPIQ test and it failed with distortion detected.

Looks like the A/P image may be slightly distorted though all of the intermediates have this “ghost” on the edges.

Solution:

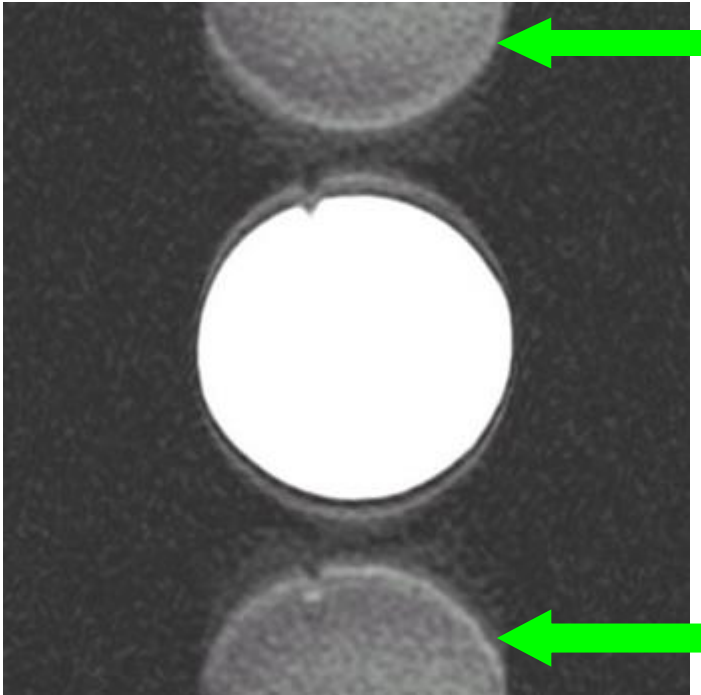
Ultimately resolved this issue using the Grafidy3 program, which at the time was not part of 9.0 software. One additional side note, after Grafidy3 was run, DWEPIQ then passed with acceptable distortion.



EPI Ghosting Artifacts

Issue description:

1.5T LX ES with an LCC magnet.
Ghosting artifacts.



Root cause:

Due to environmental vibration. Some of this was due to the cold-head but some of this was also the result of an equipment room at the site, which was directly under the MR room that was full of rotating equipment.

Solution:

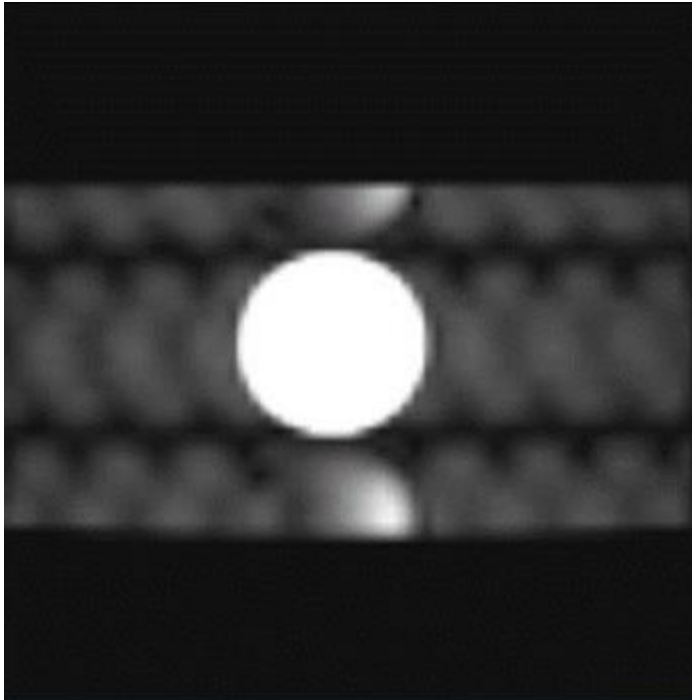
We could greatly reduce the ghost by turning off the cold-head. HSS failed on this system.

EPI Ghosting Artifacts

Issue description:

Twinspeed system.

Ghost was seen when scanning EPI with ramp sampling on. If ramp sampling was turned off, the ghost was not seen.



Root cause:

Bad Gradient filter module on the Pen Panel. FE could not find any problem with the filter through visual inspection. There were no SPT failures seen in the failure mode and no complaints of image quality problems on any other type of scans.

Solution:

Replace Gradient Filter.

EPI Ghosting Artifacts

Issue description:

1.5T LX ES ACGD gradients, and SIII magnet.
Ghosting seen on this DWEPI image

Root cause:

Due to a bad X-axis SGA (Readout gradient with this scan).

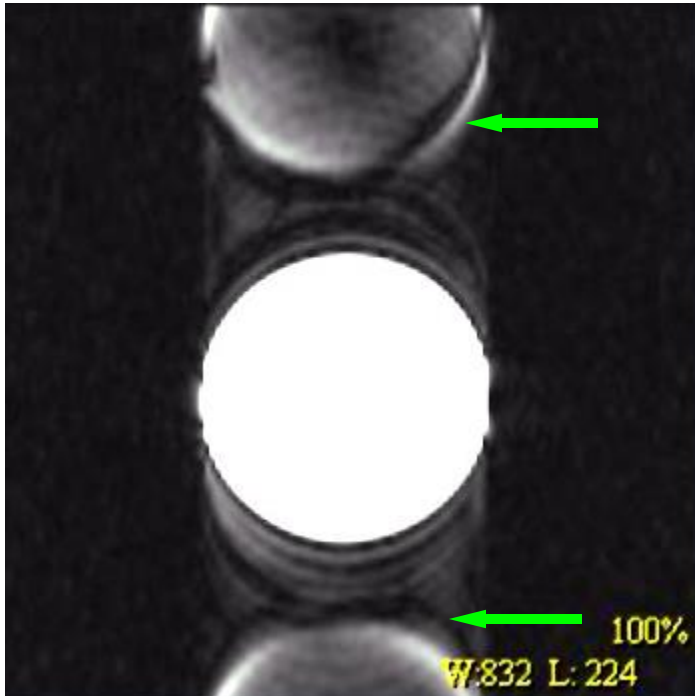
Troubleshooting:

DWEPIQ test failed with some distortion measurements.

Then Grafidy ran numerous times but it did not have any impact on this ghost.

Solution:

X-axis SGA replaced, the FE ran DWEPIQ again and found that it now passed without the distortion seen earlier.



EPI Ghosting Artifacts

Issue description:

1.5T LX ES with LCC wide-open enclosure, ACGD gradients.

Ghost seen during installation of the system.

Root cause:

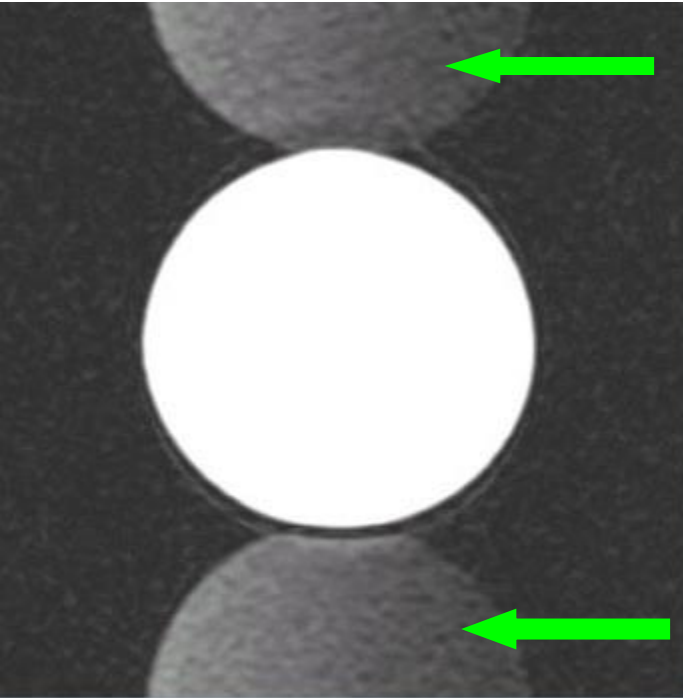
Caused by bridge vibration.

Troubleshooting:

The bridge was not properly supported on the front with the new bridge support; it was basically floating at the front of the magnet. Some notes: B0 image failed severely on BODY, whereas HEAD seemed to be unaffected by this. SPT stability looked fine. Since this was a new install, no real patients had been scanned yet so we do not know how this may have affected patient images. This may not have been as much of a problem with a real patient, which would add some weight on the bridge.

Solution:

Check that the bridge bolts are tight.

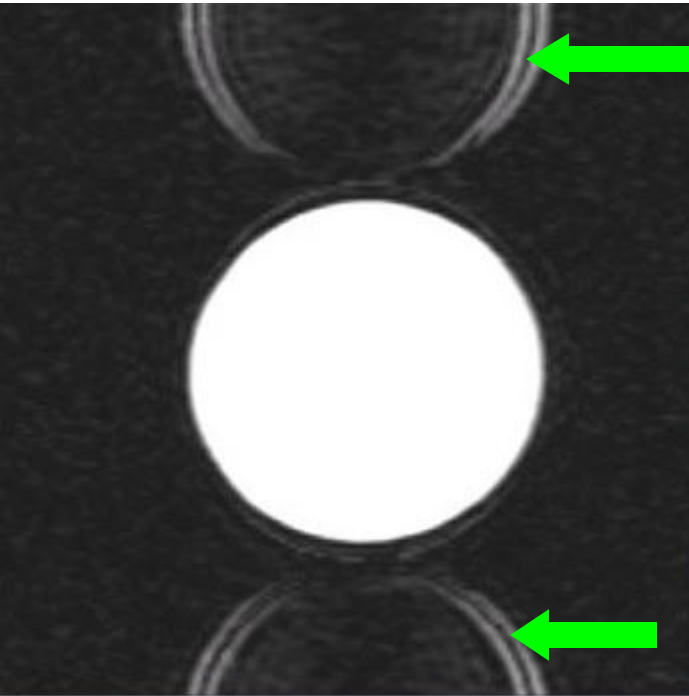


EPI Ghosting Artifacts

Issue description:

1.5T LX HS with ACGD gradients.

DWEPI scans show heavy edge ghost. The T2 image shows a heavy ghost as well.



Troubleshooting:

FE run EPT to confirm the RF shield was not shorted, B0 dither cal checks fine, and also to redo the delay values.

EPIWP has severe failures but it looks like it was failing before.

Solution:

This problem was resolved from the customer standpoint by editing the delay.dat file delay for the X-axis (X is readout for the scan they are doing). The standard delay for the X-axis generated by EPT was at 32 us, we vi edited the delay.dat file and set this delay to 23 us as seen in the other system examples and this pretty much cleared up the issue and the customer was happy. Not sure why this value was high. FE ran EPT again and it came up with 32 us again so we decided to just stick with the manually entered value at this time.

EPI Ghosting Artifacts

Issue description:

TwinSpeed system.

Combined DWEPI image in the ZOOM mode. Increased ghost and the horizontal null in the anterior ghost.

Root cause:

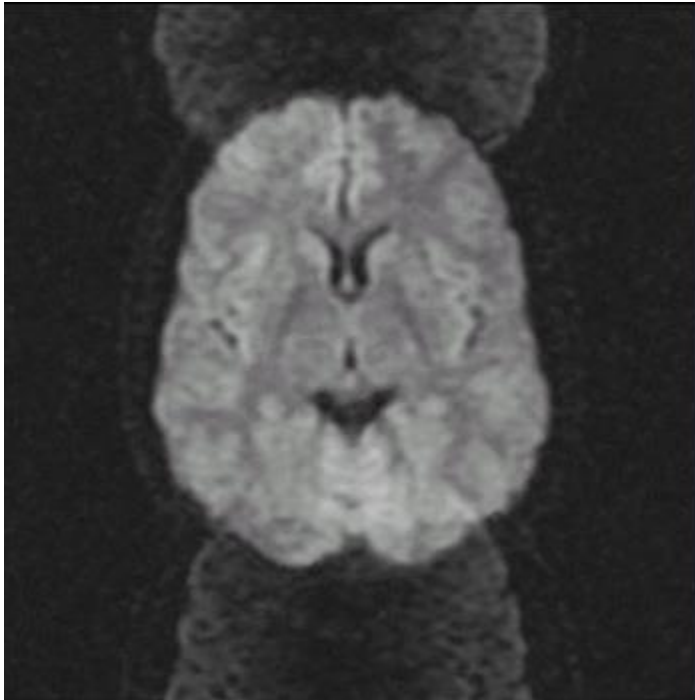
Result of no short-term eddy current compensation. The horizontal null seen in the ghost is pretty typical of missing short-term eddy comp.

Solution:

Run Grafidy and adjust short-term eddy current.



EPI Ghosting Artifacts



Issue description:

1.5T LX ES.

Ghost occurred whenever the customer ran an EPI scan with ramp sampling ON (using the fast receiver).

Ramp sampling turned off , made ghosting acceptable.

Troubleshooting:

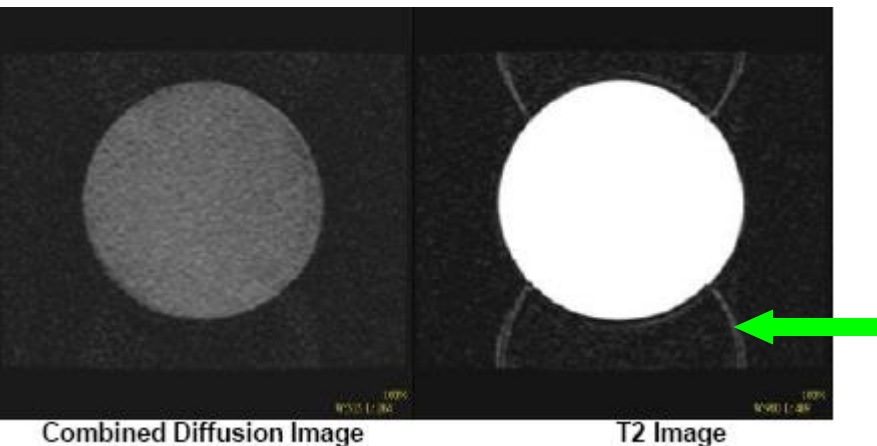
Fast receiver replaced and with no change.

Root cause:

Problem solved by swapping the ACGD control boards between the X and Y SGAs. Most probably, the problem was with the X-axis, since it was playing the readout role on these scans and when the bad control board was on the axis with the phase role, it did not bring out the problem. Either way, the problem was with an SGA. Notice that there is a vertical null in the ghost. See some additional DWEPI scans on a phantom.

Solution:

Replace X SGA.



EPI Ghosting Artifacts

Issue description:

1.5T LX HS with 8645/Gram gradient drivers.

Curved sat band through the image and increased ghost level.

Root cause:

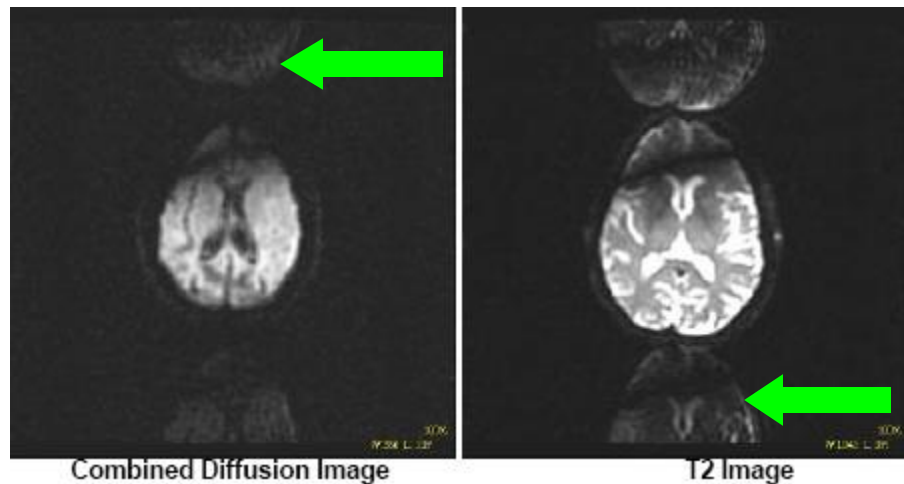
Result of a shorted RF shield.

Troubleshooting:

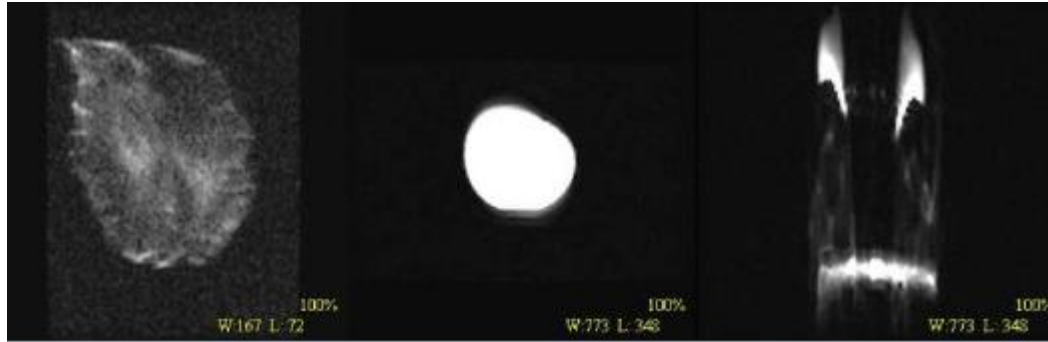
The evidence on this was seen in the B0 dither offsets after running EPT. The shield was shorted on Y and Z. The offsets were not as high as we have typically seen with a shorted shield, they were only in the 30 range.

Solution:

Replace the BRM (or the RF shield).



EPI and DWEPI Image Distortion



Issue description:

1.5T HS with 8645/Gram gradients and an SIII magnet.

Distortion to varying degrees. The T2 and the diffusion images are affected the same. EPI phantom shows simply geometry distortion and the images of patients show what looks like extreme stretching of the images.

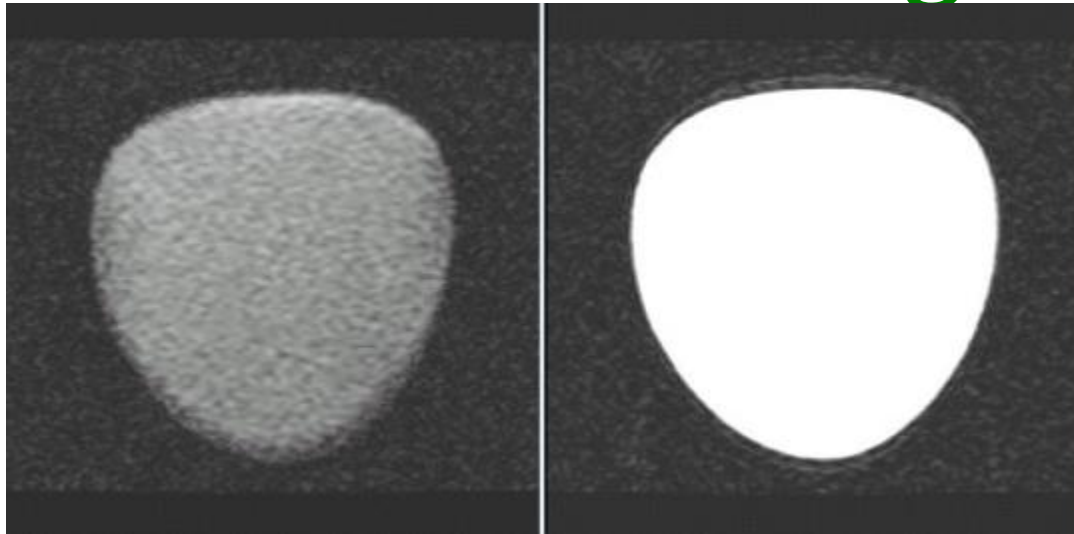
Root cause:

Variable bandwidth of 16 KHz. Too narrow receive BW with EPI will cause distortion. Part of this extreme distortion is due to autoshim trying to compensate for the BW problem. That is why the patient images are more distorted than the EPI phantom image. The phantom diameter is smaller than the autoshim FOV default so it fails and leaves the Grad shim values as default. As a result, autoshim does not have an affect on the phantom scan.

Solution:

Increase bandwidth value: 125 KHz, is the most used value.

EPI and DWEPI Image Distorsion



Issue description:

1.0T HS LX mobile with SGD gradients.

Distortion in these images.

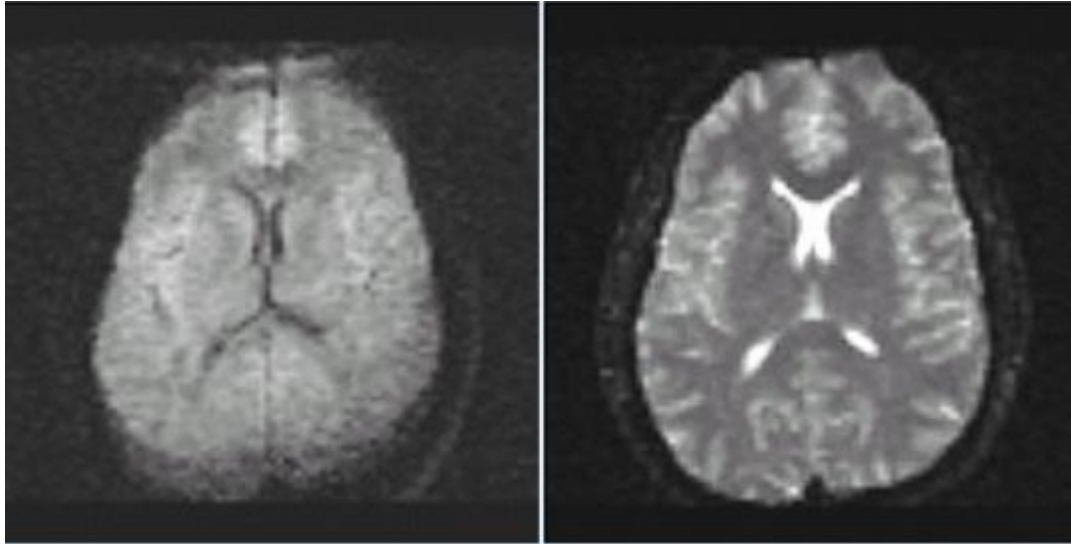
Troubleshooting:

Round phantom scanned. This was caused by a very poor magnet shim. The actual shim file could not be obtained but it was noted that the overall shim over 45 cm with a 500 bandwidth was 98 Hz. Worst harmonics were X2-Y2=-274, ZY=150, X=200, and Y=-200.

Solution:

To correct this problem, a supercon shim had to be performed. Note that the T2 image is distorted and the diffusion image is distorted in the same way. This indicates that the distortion is in the EPI image and not caused by the diffusion pulse. This typically suggests that something like eddy currents are not likely to cause this distortion.

EPI and DWEPI Image Distortion



Issue description:

1.5T LX ES with 8645/Gram gradient drivers and an SIV magnet.
Distortion in the combined diffusion image.

Root cause:

This distortion was by uncompensated eddy currents.

Solution:

Run Grafidy.

EPI and DWEPI Image Distortion

Issue description:

1.5T HS Excite.

Axial DWEPI image with an 8-channel NVARRAY coil in the 8NVHEAD_A configuration.

Root cause:

Distortion in the image turned out to be caused by a bad NVARRAY coil.

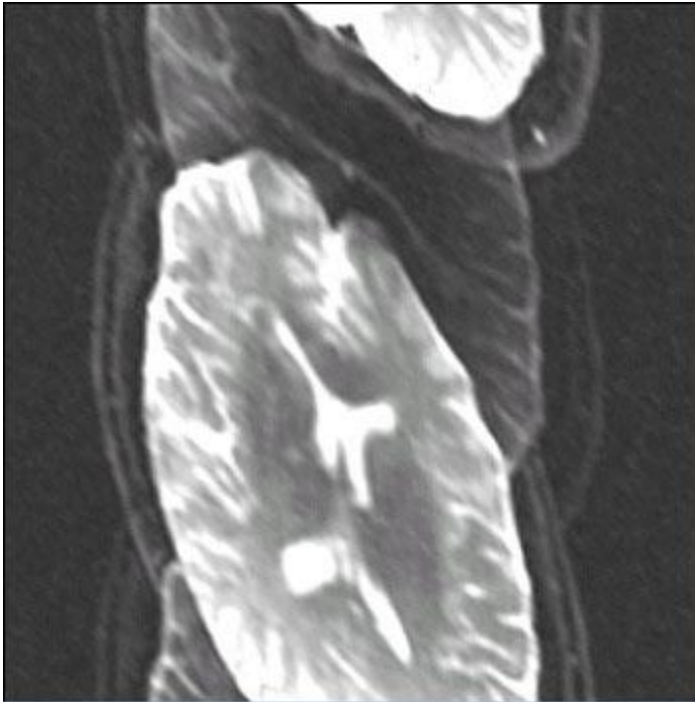
Troubleshooting:

DWEPI scan done with the 8HRBRAIN coil and found that the image was not distorted. Additionally, this coil had just been recently for a different problem which made the coil suspect.

Old NVARRAY was still on site, DWEPI scan done with it without distortion. This confirmed that the problem was the coil.

Solution:

Ordered a new 8-channel NVARRAY coil.



EPI and DWEPI Image Distorsion

Issue description:

1.5T 5.X HS.

Brain exam with a DWEPI sequence.

Severe distortion reproducible every time. T2 image and combined diffusion image distorted the same way. Patient images appear “stretched”.

Troubleshooting:

Scan on EPI phantom done and images now distort as smashed in the A/P direction. Both the diffusion and the T2 images (EPI without diffusion applied) are distorted in the same way.

This is not as likely to be caused by eddy currents since the T2 is also distorted.

Root cause:

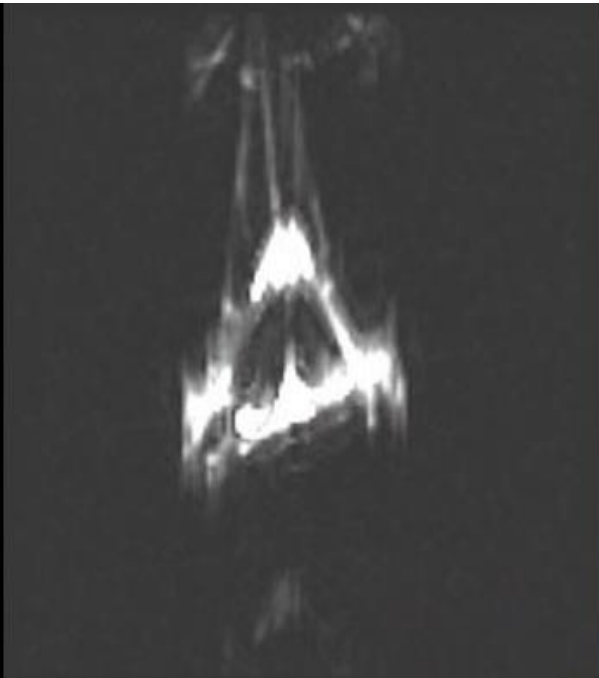
Variable BW and set the receive bandwidth to 16K.

Too narrow receive bandwidth will cause geometry distortion with EPI.

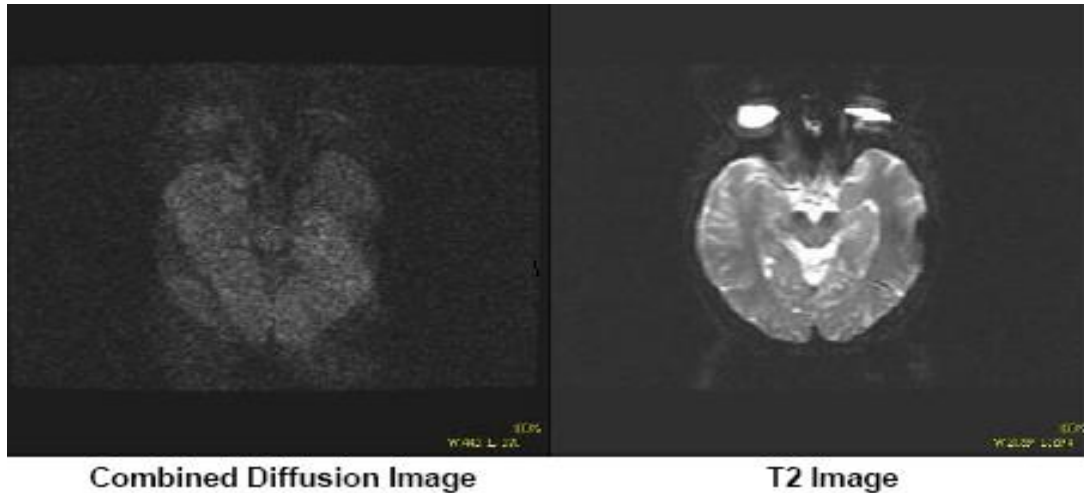
You may wonder why the images on the phantom are distorted differently. The reason why is that the EPI phantom FOV is less than the autoshim FOV setting which causes autoshim to fail and therefore defaults to the saved Grad shim values. On the patient scan, autoshim did execute and this is what was causing the images to be “stretched” as autoshim was incorrectly trying to correct for the distortion due to the BW issue.

Solution:

Increase bandwidth value: 125KHz, is the most used



EPI Wormhole Artifacts and Signal Void



Issue description:

1.5T LX ES.

Root cause:

Caused by a poor magnet shim. Note the problem is seen primarily on the diffusion images, while no real great effect is seen on the T2 (EPI) images.

Solution:

Run supercon shim.

EPI Wormhole Artifacts and Signal Void

Issue description:

1.5T SmartSpeed.

Wormhole artifact at the base of the skull as displayed in the combined diffusion system image.

Root cause:

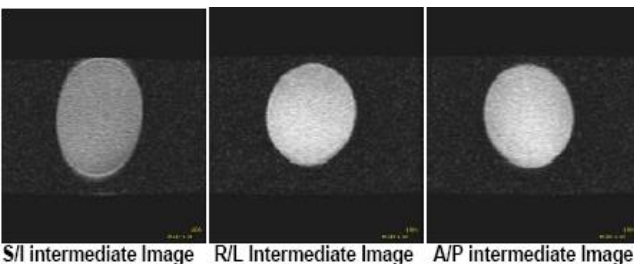
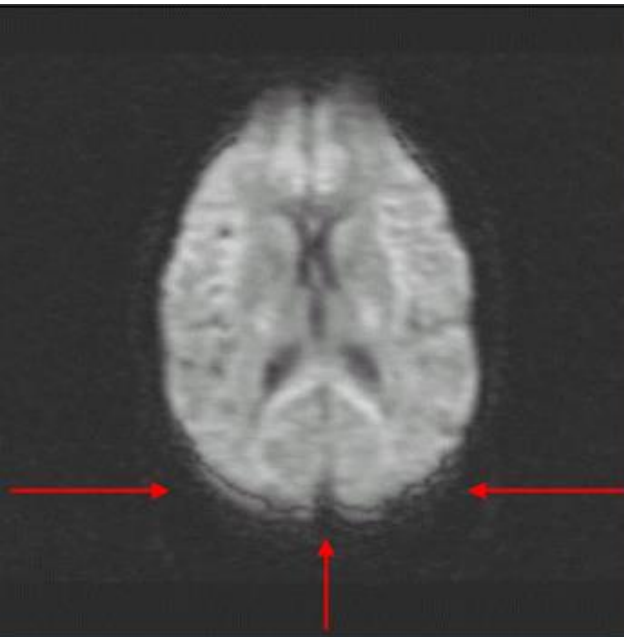
Eddy current problem, which caused varying degrees of distortion on the intermediate diffusion images. This distortion causes a mismatch of the edges of the three intermediate diffusion images when put together to make the combined diffusion image, which the customer saw.

Troubleshooting:

To prove this out, a simple experiment was conducted with the EPI phantom. The customer's DWEPI protocol was used to enable "recon all" from the diffusion options page in the scan setup. These should be images of a round phantom, but note the primary issue is on the S/I diffusion images, it looks like an egg.

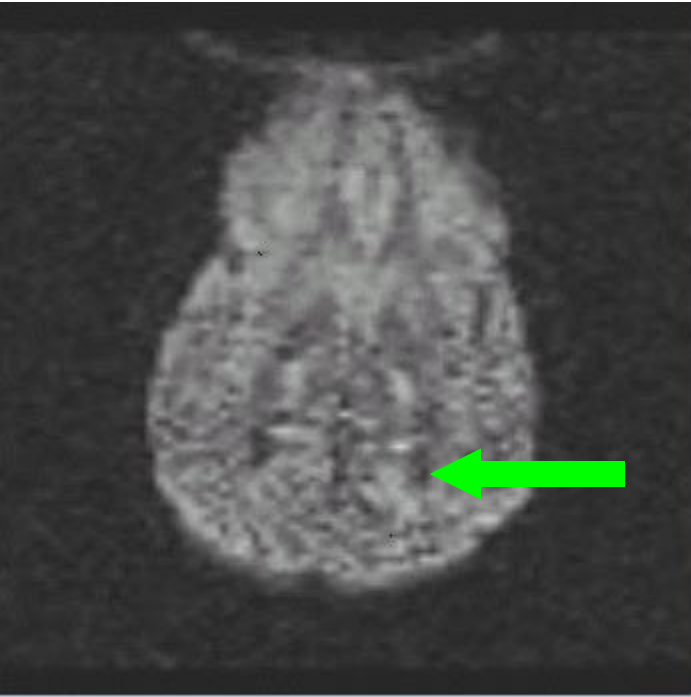
Solution:

Run Grafidy to fix this problem. This distortion is due to uncompensated eddy currents.



S/I Intermediate Image R/L Intermediate Image A/P Intermediate Image

EPI Wormhole Artifacts and Signal Void



Issue description:

1.0T HS with 8645/Gram gradient drivers.

Wormhole artifacts in combined diffusion images.

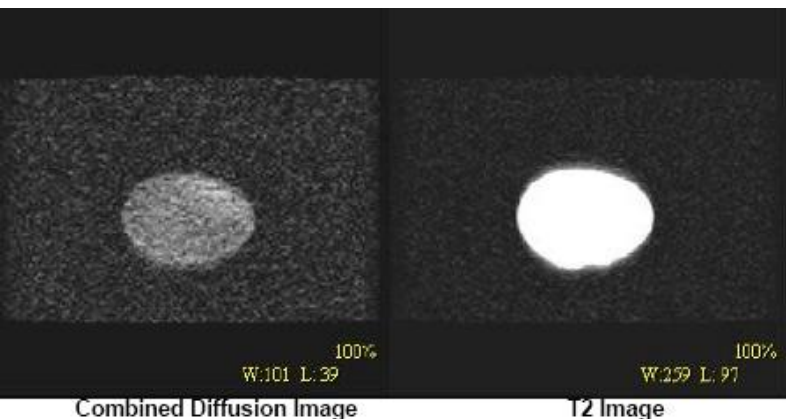
Troubleshooting:

In order to see the intermediate diffusion images, a test with “recon all” was performed. The A/P intermediate diffusion image had signal void.

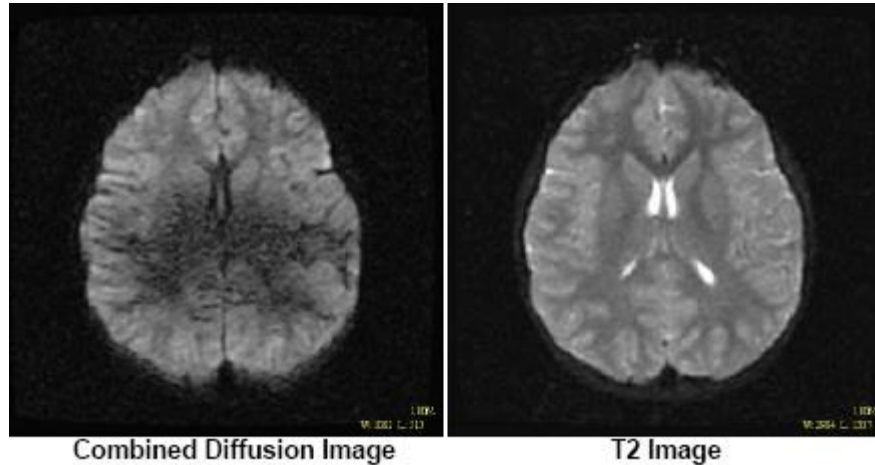
In the log we find Y-axis GPM short overloads when the customer scans these diffusion scans. These short overloads do not cause the scan to stop, but were in fact the cause of these wormhole artifacts. What we did to prove this out was to swap the Y and Z axis 8645 power modules and then ran the test scan again with “recon all”. We found that the power module errors now moved to Z-axis, and additionally the S/I intermediate diffusion image now had signal void and the A/P intermediate was now good.

Solution:

Replace the Y-axis 8645 Power Module.



EPI Wormhole Artifacts and Signal Void



Issue description:

TwinSpeed system.

DWEPI scan in the ZOOM mode. The signal void and wormhole artifacts on the combined diffusion image on the left were a result of the gradient coil rocking due to loose upper radial supports.

Troubleshooting:

To better confirm this as the likely cause, scan this again as “geservice” and turn on “recon all” under diffusion options in the scan setup. This is done to recon all of the intermediate diffusion images. You will see the same wormholes and signal void on the R/L diffusion image since this is the direction the coil is allowed to rock when the radial supports are loose.

One additional note, you will not be able to reproduce wormhole artifacts on a phantom.

Solution:

Check and tighten BRM support bolts.

EPI Wormhole Artifacts and Signal Void

Issue description:

1.5T LX ES with SGD gradients and an Erbtec RF amp. Signal void. This problem occurs when the customers use the NVARRAY coil, which is what they typically use when running this DWEPI scan. This was not intermittent.

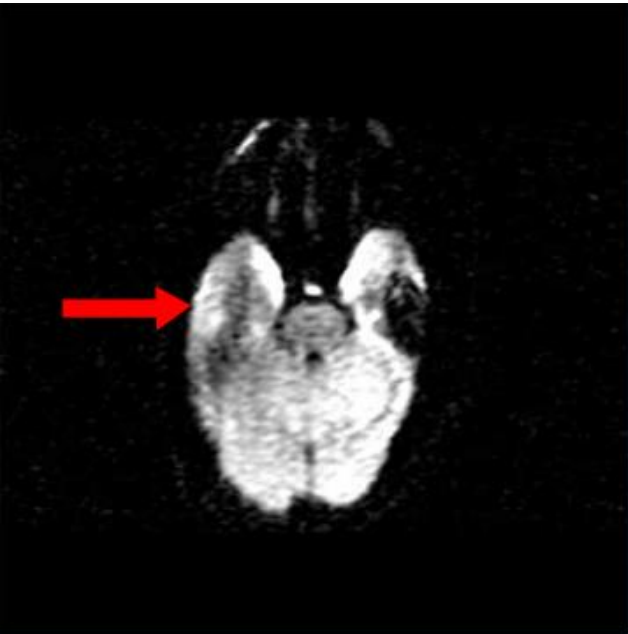
Troubleshooting:

Scan performed with “recon all” selected under the diffusion option in order to see all of the intermediate diffusion images. Three of the intermediate diffusion images had this signal void. Same sequence ran with the HEAD coil and not see the problem.

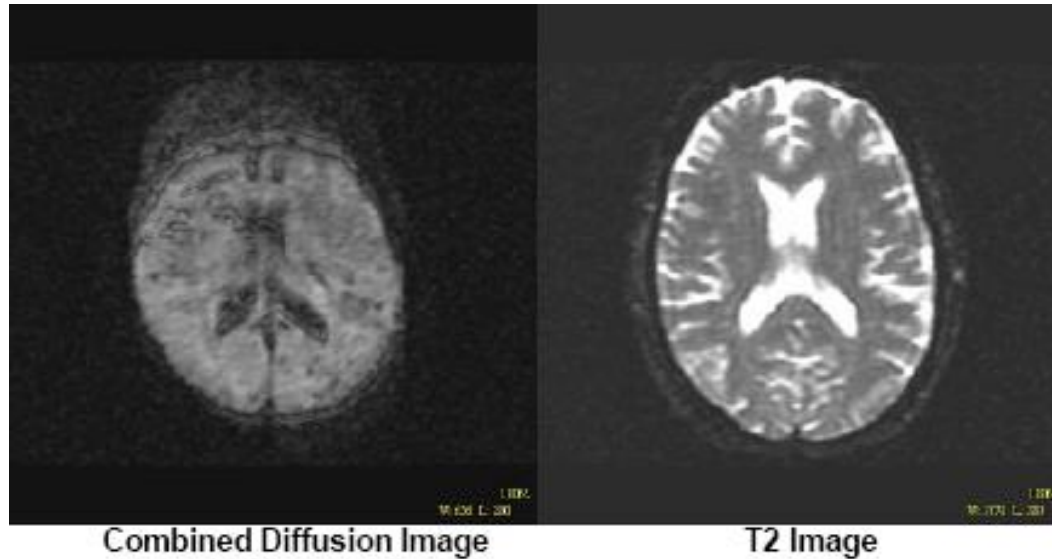
Note that the NV Array is a surface coil so it is Body transmit. With this difference decided to try the same scan with the body coil. Found that the body coil scan also exhibited the same signal loss. With this started looking at a body transmit problem. Ultimately found that the body RF output out of the Erbtec was clipping at about TG=140 (the TG for the NV Array scan usually ended up around 145-150).

Solution:

Erbtec amp replaced to fix this. The customer was not complaining about any other IQ issues at the time but after replacing the Erbtec amp, they noticed the spine work was improved as well. 80



EPI Wormhole Artifacts and Signal Void



Issue description:

1.0T LX HS with an LCC magnet.

IQ issue in combined diffusion image caused by a magnet shim problem. Severe distortion as well as some wormhole artifacts was also caused. T2 image in the right; not affected by this shim problem like the diffusion image.

Solution:

Run supercon shim.

EPI Wormhole Artifacts and Signal Void

Issue description:

1.5T ES.

IQ issue on combined diffusion image.

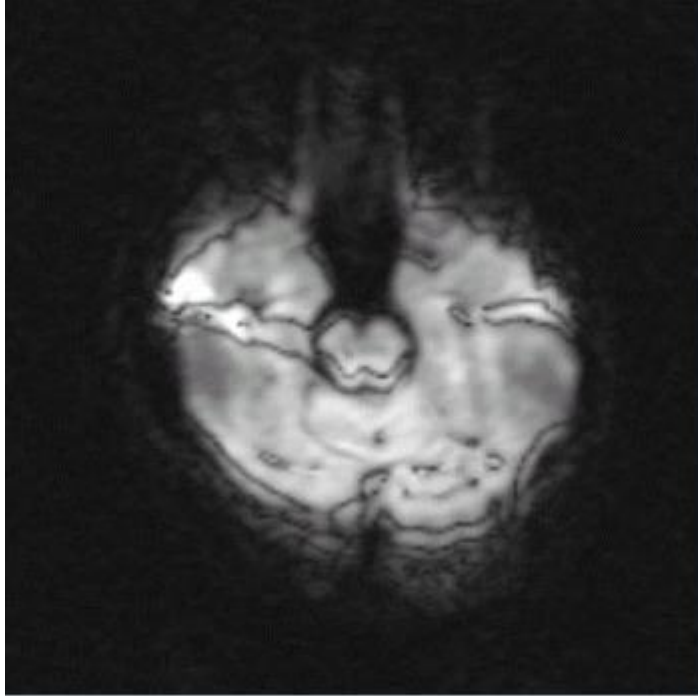
Root cause:

Caused by two problems.

1. Geometry distortion. FE had neglected to run cross term eddy current comp. No crossterm eddy currents caused the distortion.
2. Wormhole artifacts were caused by loose BRM upper radial supports, which allowed the gradient coil to “rock” during the DWEPI scan.

Solution:

Check BRM support tightness and run Grafidy.



Slice to Slice Intensity Variation and Curved Sat Band

Issue description:

1.5T Twinspeed Excite.

DWEPI scan done in ZOOM mode.

Shaded band and increased ghost.

Root cause:

Result of no short-term eddy currents. This was an oversight by the FE when he ran Grafidy3. He did not enable the short-term defaults.

Solution:

Go back into Grafidy3 and enable the short-term defaults for both WHOLE and ZOOM and then run EPT again on both WHOLE and ZOOM.

Slice to Slice Intensity Variation and Curved Sat Band

Issue description:

1.5T LX mobile system.

Curved sat band, which is running through the EPI image artifact example.

Note: This is the DQA phantom; the horizontal black band through the image is the phantom.

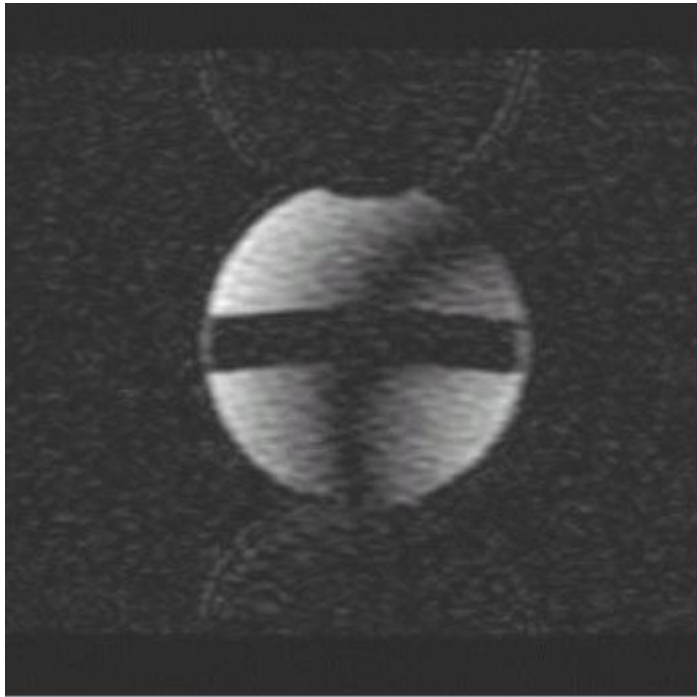
There is also slightly increased ghost.

Root cause:

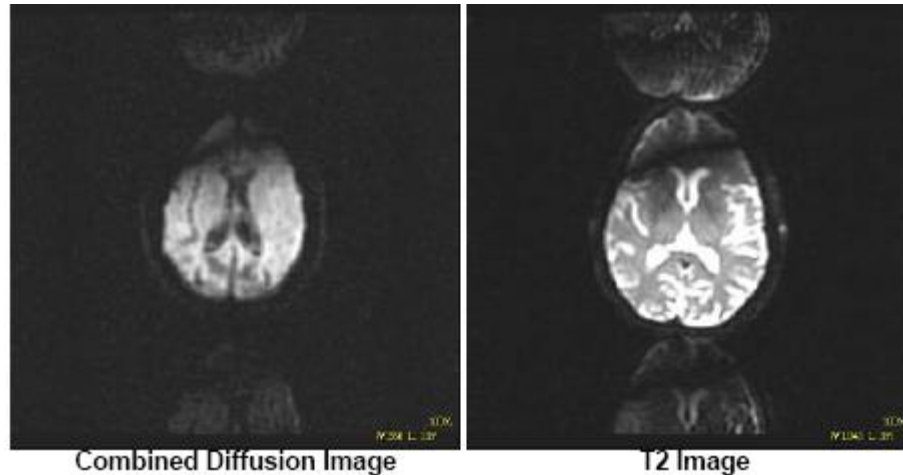
Due to a shorted RF shield in the BRM. Evidence of this was seen in the B0 dither offsets. The Z offsets were up in the 80s. The shorted RF shield allows very short-term B0 eddy currents to cause timing problems with the EPI spectral spatial pulse, which is what is causing this curved sat band.

Solution:

Replace RF shield (or BRM).



Slice to Slice Intensity Variation and Curved Sat Band



Issue description:

1.5T LX HS with 8645/Gram gradient drivers.

Curved sat band through the image and increased ghost level.

Root cause:

Result of a shorted RF shield. The evidence on this was seen in the b0 dither offsets after running EPT. The shield was shorted on Y and Z. The offsets were not as high as we have typically seen with a shorted shield, they were only in the 30 range.

Solution:

The fix was to replace the BRM.

Slice to Slice Intensity Variation and Curved Sat Band

Issue description:

TwinSpeed system.

Entire series of EPI images done in ZOOM mode with the HEAD coil. Slices at the beginning and the end of the scan range saturated and have fewer signals.

Root cause:

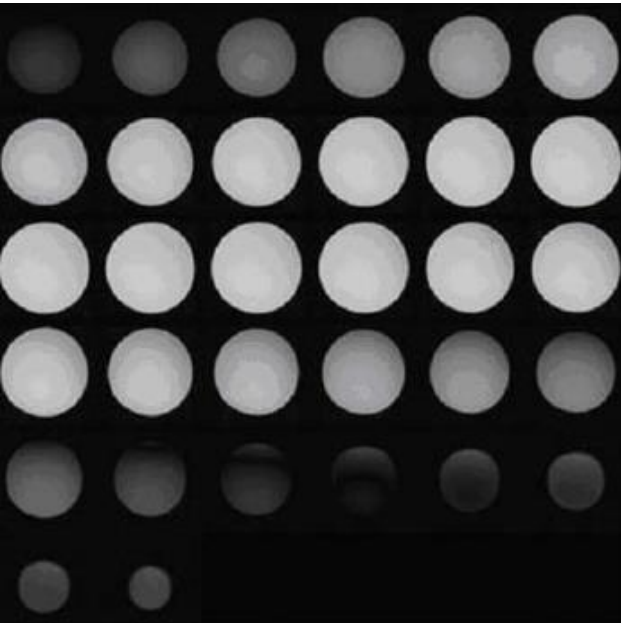
Result of misalignment of the EPI spectral spatial pulse.

Troubleshooting:

To prove this, perform a simple test by scanning the EPI sequence with FatSat turned on. When FatSat is on, the spectral spatial pulse is turned off. If the outer slices are not saturated with FatSat on, then conclude that the outer slices are saturated by the EPI spectral spatial pulse. Misalignment can be caused by a couple of different things; one is the incorrect value for `psdRfWait` in the `GradientConfig.cfg` file. It appears that the default for ZOOM mode with 9.0 and 9.1 software may not be optimal for some systems. The default value is set at 174 us in ZOOM mode. We ran a prototype calibration, which calculates this value and found the more optimal value to be 168 us.

Solution:

We modified the `psdRfWait1.5T` value in the `GradientConfig.cfg.ZOOM` file from 174 us to 168 us and scanned the same sequence again.



Other EPI Artifacts

Issue description:

1.5T LX ES.

Artifact on the T2 image and on the combined diffusion image. So, this is an EPI problem (not the result of applying diffusion). The scan sequence was using ramp sampling, which the fast receiver is used when doing EPI with ramp sampling.

Root cause:

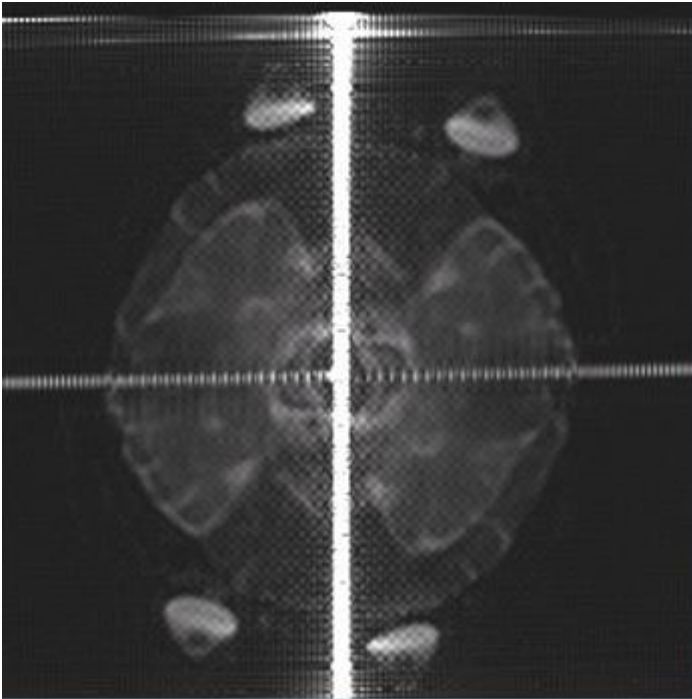
The fast receiver caused the artifact.

Troubleshooting:

Ramp sampling turned off, which put the system back to the CERD receiver zero, and the artifact went away.

Solution:

Fast receiver replaced.

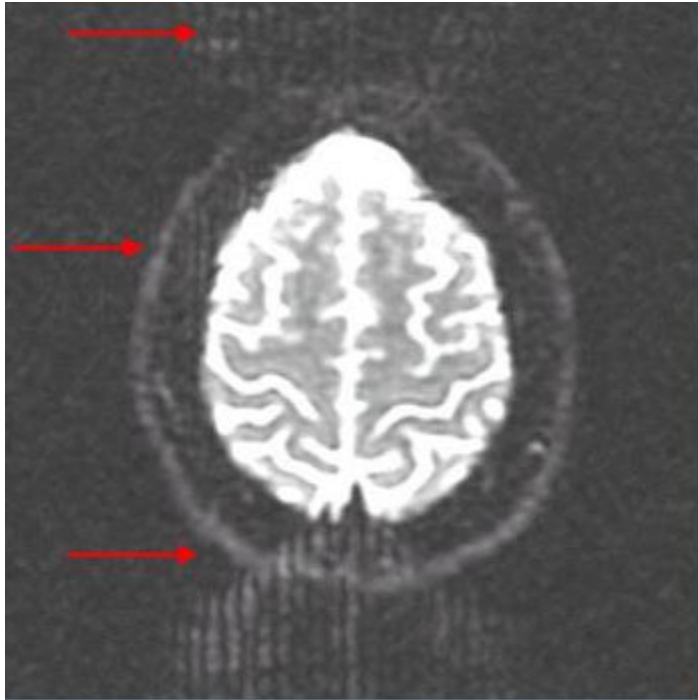


Other EPI Artifacts

Issue description:

1.5T HS Excite.

DWEPI scan with the 8HRBRAIN coil. Increased ghost on this image and more importantly the structured “lines” in the ghost most noticeable in the ghost than in the image.



Troubleshooting:

Same sequence scanned with standard HEAD coil, no “lines” and no ghost. Replacing the 8-channel Brain coil, not resolved the issue. This problem turned out to be caused by a problem with the EPI bandpass asymmetry calibration files. BACC was run when this system was upgraded to Excite but evidently something did not calibrate correctly.

Solution:

Simply run BACC again.

PROPELLER Artifacts

Crinkling Artifacts

Blurring

Signal Void

ADC (Apparent Diffusion Coefficient) & eADC Maps Issues

Ripples

RF Leak and White Pixels

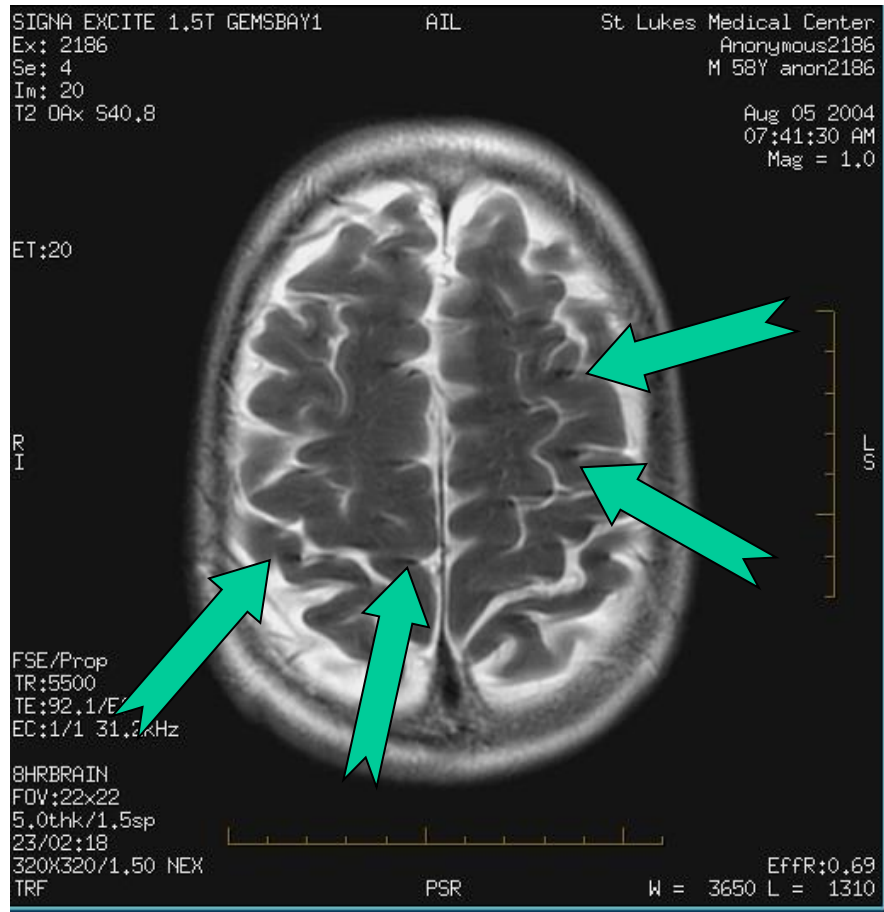
Lines criss-crossing the image

Contrast Issue For Low TR in Research Mode

Crinkling

- Crinkling can be seen in either T2 or T2 FLAIR Propeller, and DW Propeller in the case of patient motion
- Crinkling is related to:
 - The protocol...sub-optimal combinations of Frequency and ETL
 - The amount of patient motion
 - The anatomy...more common in superior slices

Crinkling from sub-optimal protocol



•Issue Description:

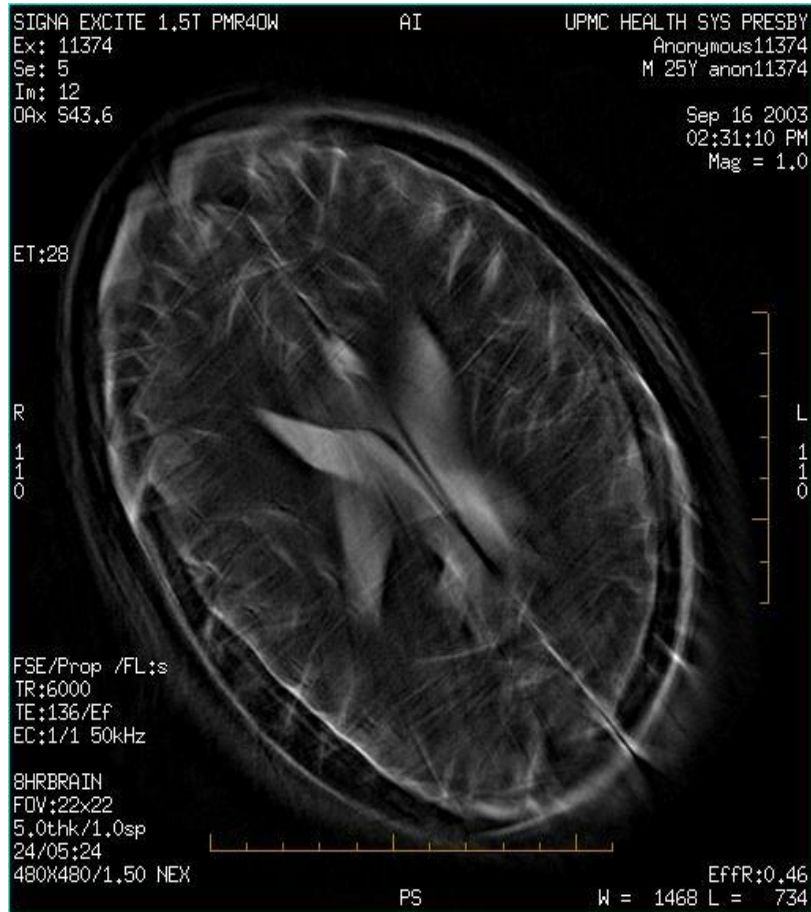
- May be described as diffuse white whiffs from sharp edges or black streaks
- Seen in superior slices of non-moving patients
- More likely in large Frequency/low ETL prescriptions...in this example the minimum ETL (20) has been prescribed

•Solution:

- Increase ETL relative to Frequency

Affects: T2 & T2 Flair Propeller

Crinkling from extreme patient motion



•Issue Description:

- Most likely to be described as diffuse white whiffs from sharp edges
- Head may be captured at an angle

•Solution:

- Increased ETL may result in better motion reduction, though some variability in results is to be expected with extreme patient motion
- Immobilize the head

Affects: All Propeller Applications

Crinkling...Prescription Rules of Thumb

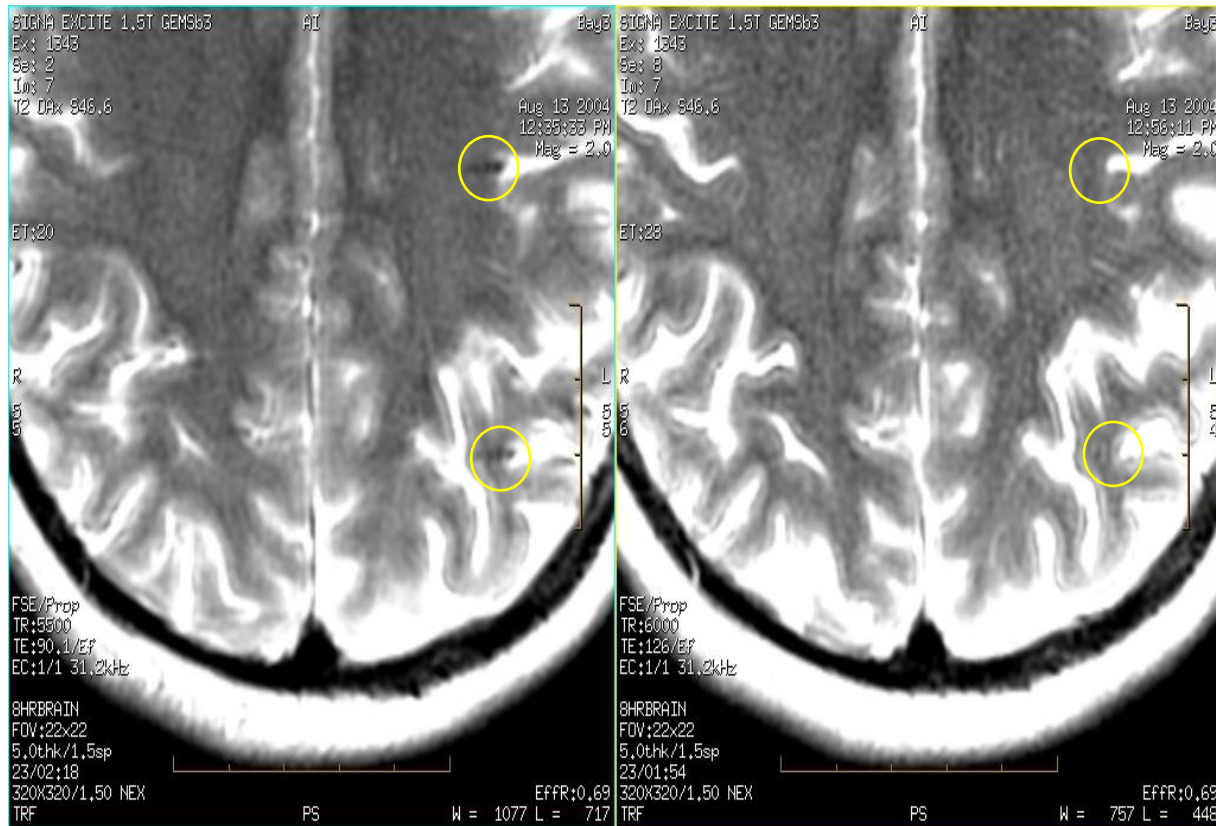
- Consider the following tradeoffs when prescribing propeller:
- As Frequency increases, the change of crinkling increases...increase ETL to reduce crinkling
- As ETL increases, TE increases...increase Bandwidth to lower TE
- As Bandwidth increases, SNR decreases

Remember that Crinkling is also affected by anatomy and the severity of patient motion.

Crinkling...artifacts are removed by using a larger ETL

ETL = 20 (TE=90)

ETL = 28 (TE=126)

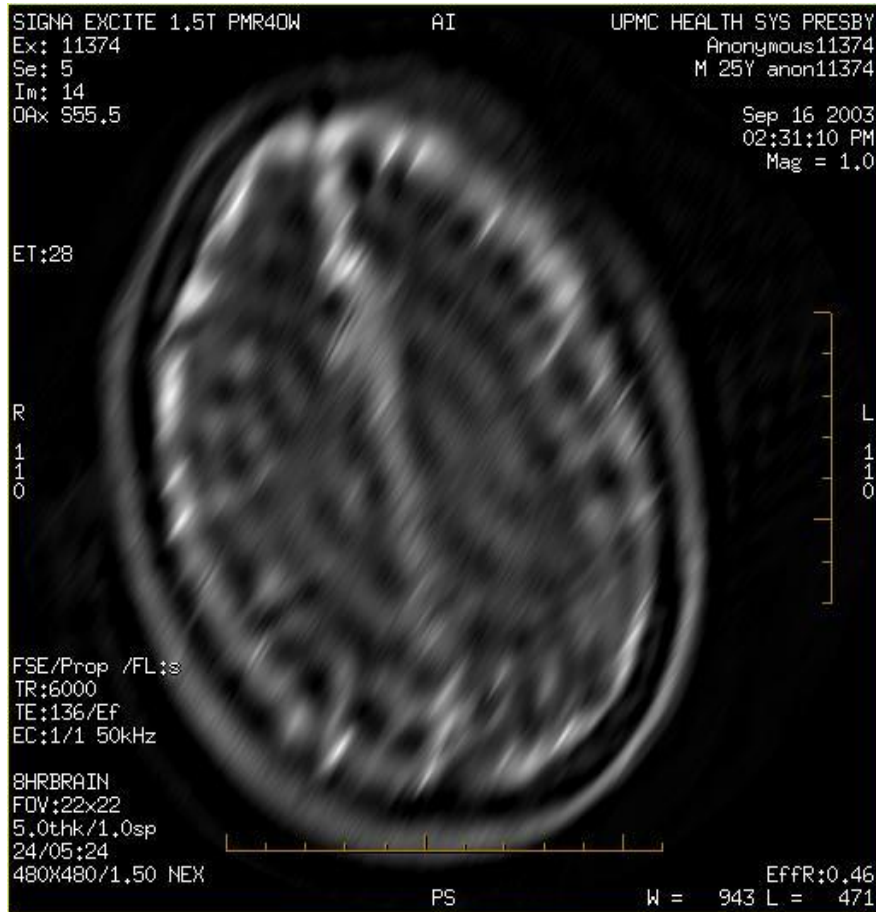


Why? ETL determines blade width. The wider the blade, the more data for blade comparison and correction.

Blurred Image

- Blurred images can be seen with all 3 Propeller applications
- Blurred images result from a failure to match enough blades... The following message will appear in the GE syslog “Image quality may be degraded due to poor blade correlation...”
- Blurred images are related to:
 - Extreme patient motion
 - In the case of DW Propeller, system issues...See Signal Voids Solution

Blurred Image



Affects: All Propeller Applications

- **Issue Description:**

- Blurred image, head may be captured at an angle
- The following message will appear in the GE syslog
“Image quality may be degraded due to poor blade correlation...”

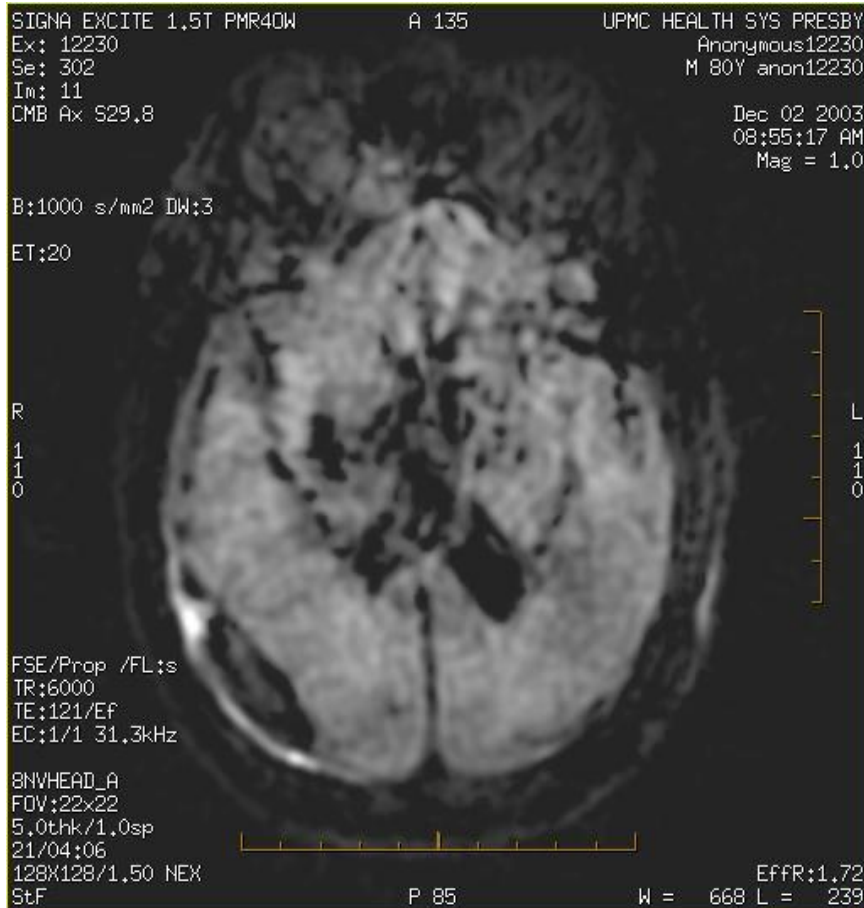
- **Solution:**

- Increased ETL may result in better blade correlation, though some variability in results is to be expected with extreme patient motion

Signal Voids

- Signal Voids are seen in the diffusion weighted images produced by DW Propeller
- Signal Voids are related to:
 - System calibrations out of spec (e.g. eddy currents)
 - Vibration of system hardware components during diffusion weighted scanning

Signal Voids

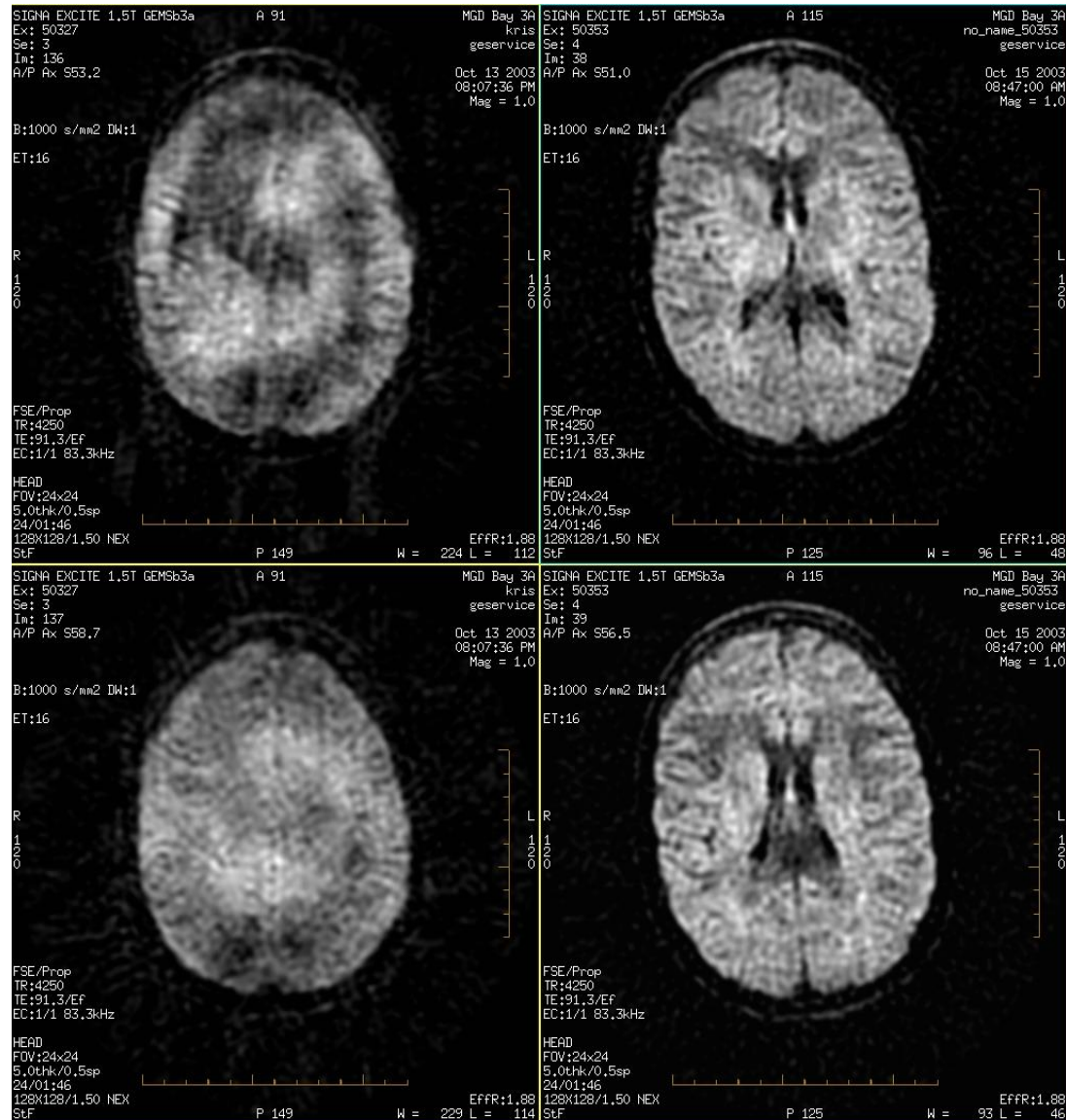


- **Issue Description:**
 - Black holes in diffusion weighted images image
 - May not occur with every scan
- **Solution:**
 - Ensure the system is in spec and that the proper bridge support is installed

Affects: DW Propeller

Signal Voids...effect of bridge support

Before
Bridge
Support

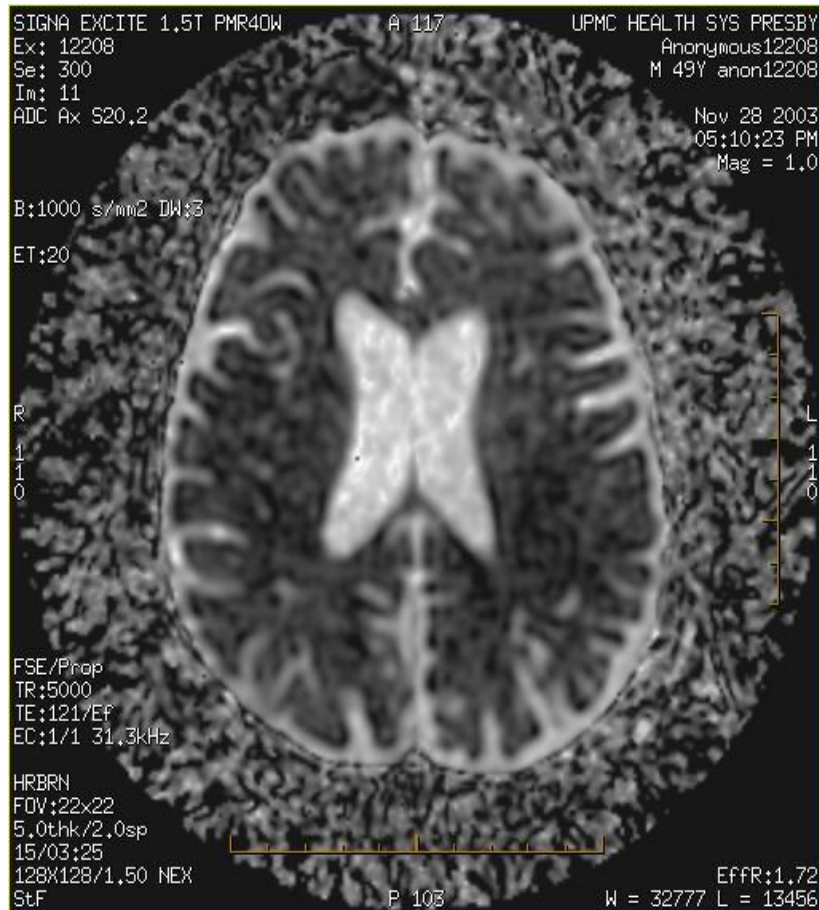


After
Bridge
Support

ADC/eADC maps show elevated noise in the background

- ADC and eADC maps automatically generated during DW Propeller image reconstruction may have more noise in the background than those generated in Functool
- Functool allows the noise to be thresholded out by the operator...the same cannot be done during image reconstruction and its hard to achieve the similar results with a fixed threshold

ADC/eADC maps show elevated noise in the background



- **Issue Description:**

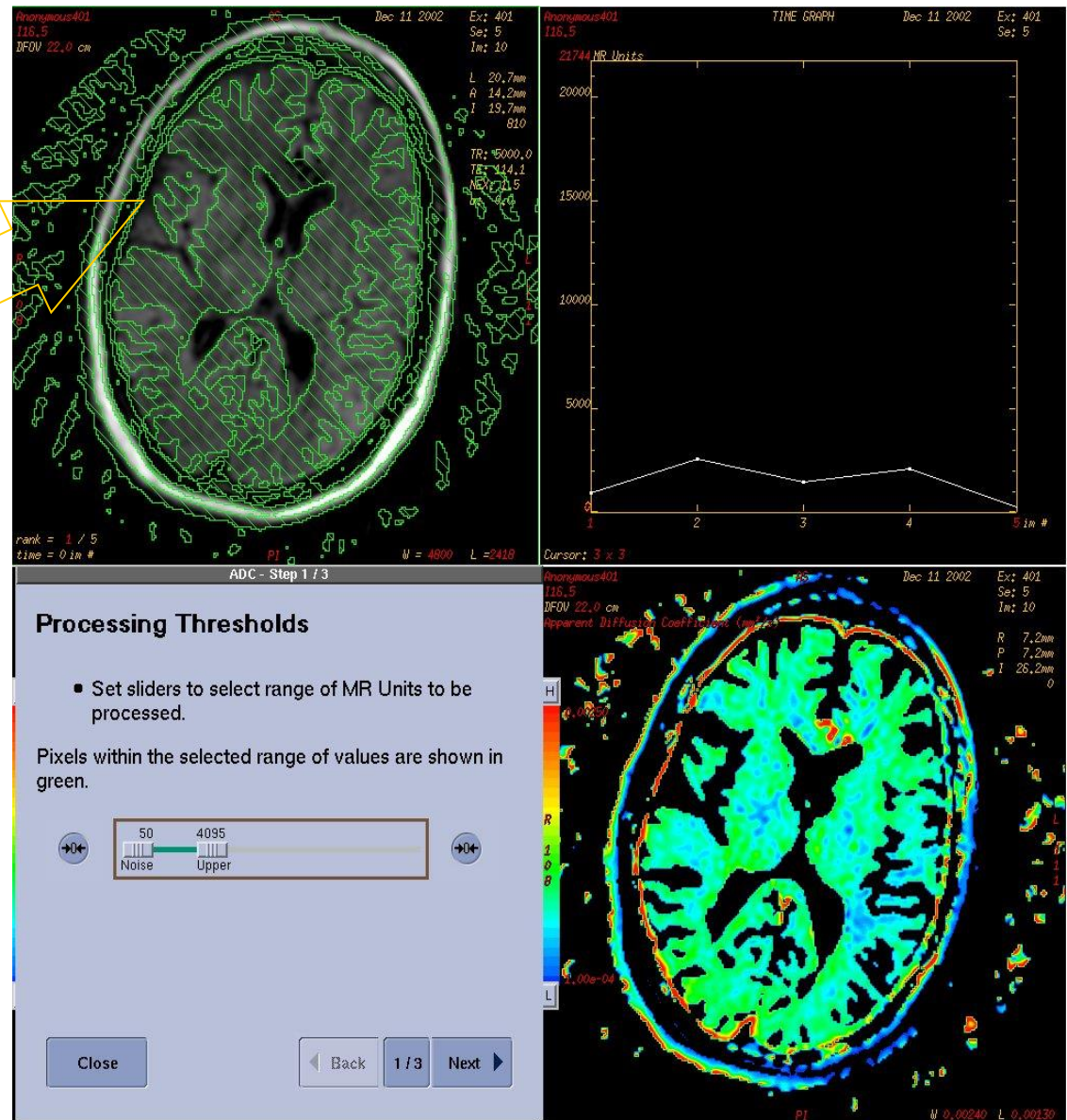
- ADC and eADC maps may have more noise in the background than Functool generated images
- There is no effect on anatomy only aesthetics

- **Solution:**

- Scan without SCIC and use Functool, which allows for operator control of the noise threshold

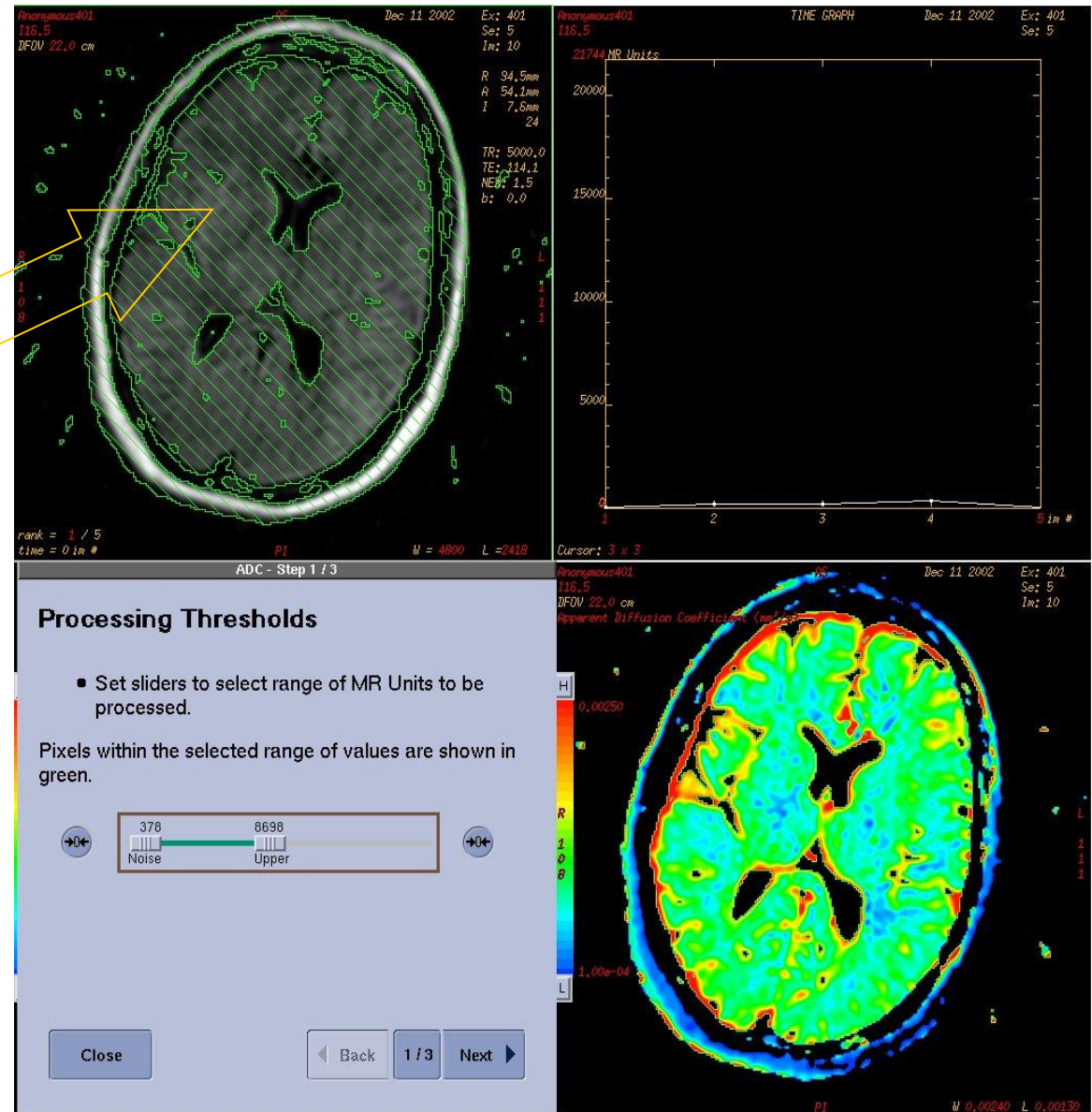
Affects: DW Propeller ADC and eADC Maps

Example: Results of using default threshold



Hard to get good results with a fixed threshold... tissue is segmented out

Example: Results of operator selected threshold

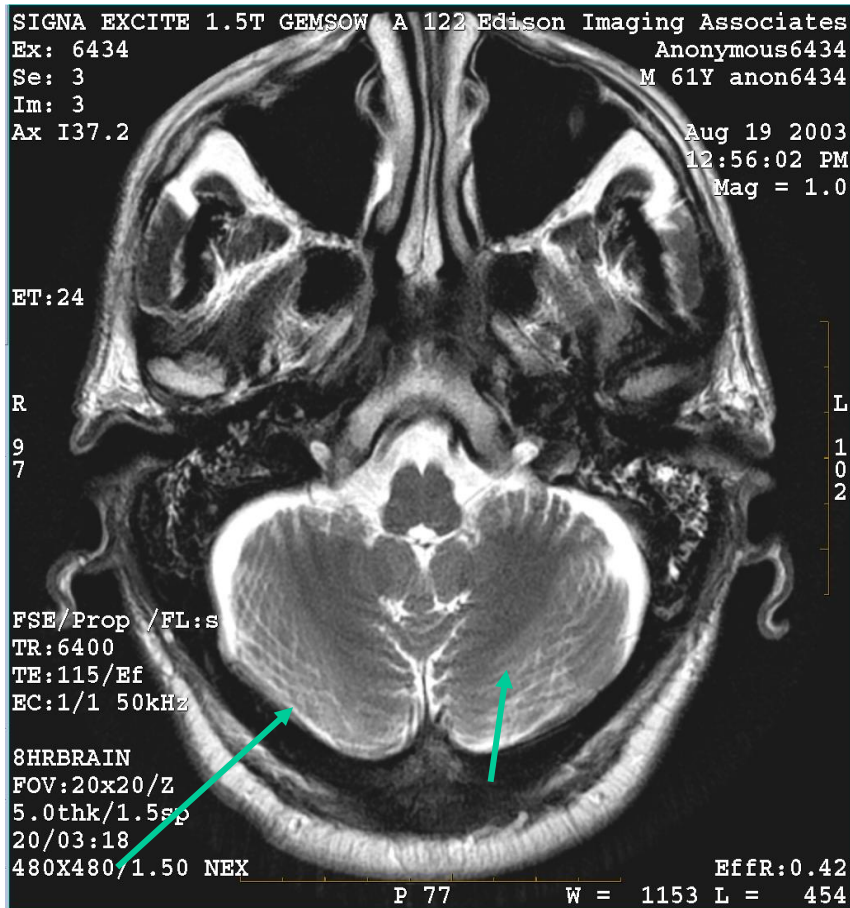


Better results
when the
operator
manually adjusts
the threshold...
tissue not
segmented out

Ripples

- Ripples can be caused by wrap and can be seen with all 3 Propeller applications
 - Wrap looks different than expected due to the radial acquisition
- One of the reasons Propeller scan plane prescription is limited to axial oblique is to avoid wrap.

Ripples

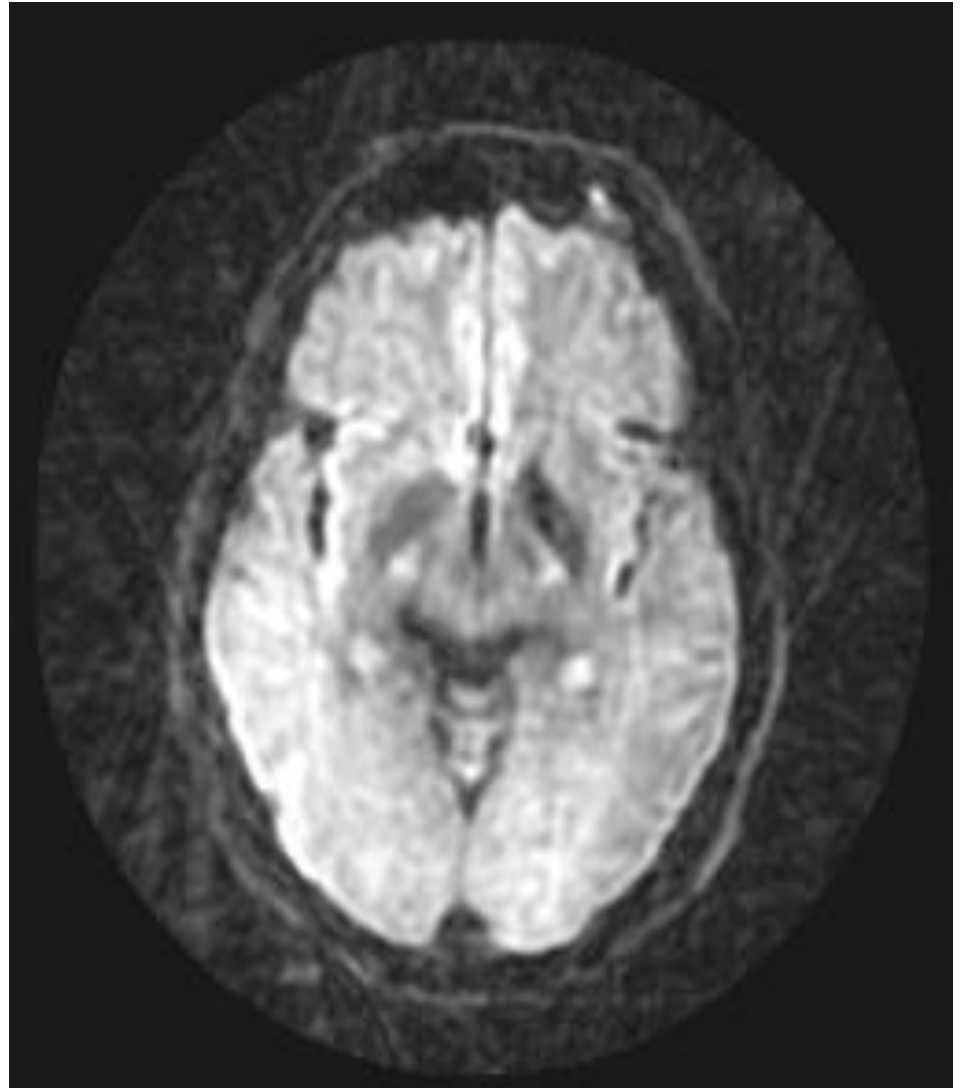


- **Issue Description:**
 - In this example, signal from the nose wraps around as radial lines or ridges
- **Solution:**
 - Increase FOV and/or reposition the slice prescription

Affects: All Propeller Applications

Ripples...another example

Affects: All Propeller Applications



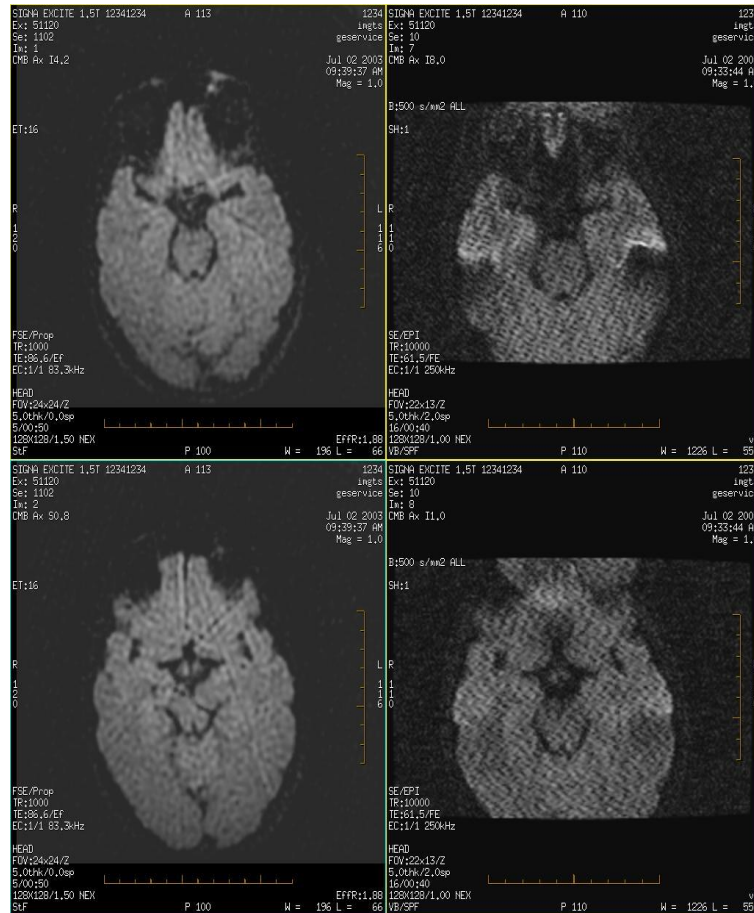
RF Leak and White Pixels

- RF Leak and White Pixel problems can be seen with all 3 Propeller applications
- Both look different than expected due to the radial acquisition

White Pixels look different with Propeller

DW
Propeller...

“criss-crossed
lines”

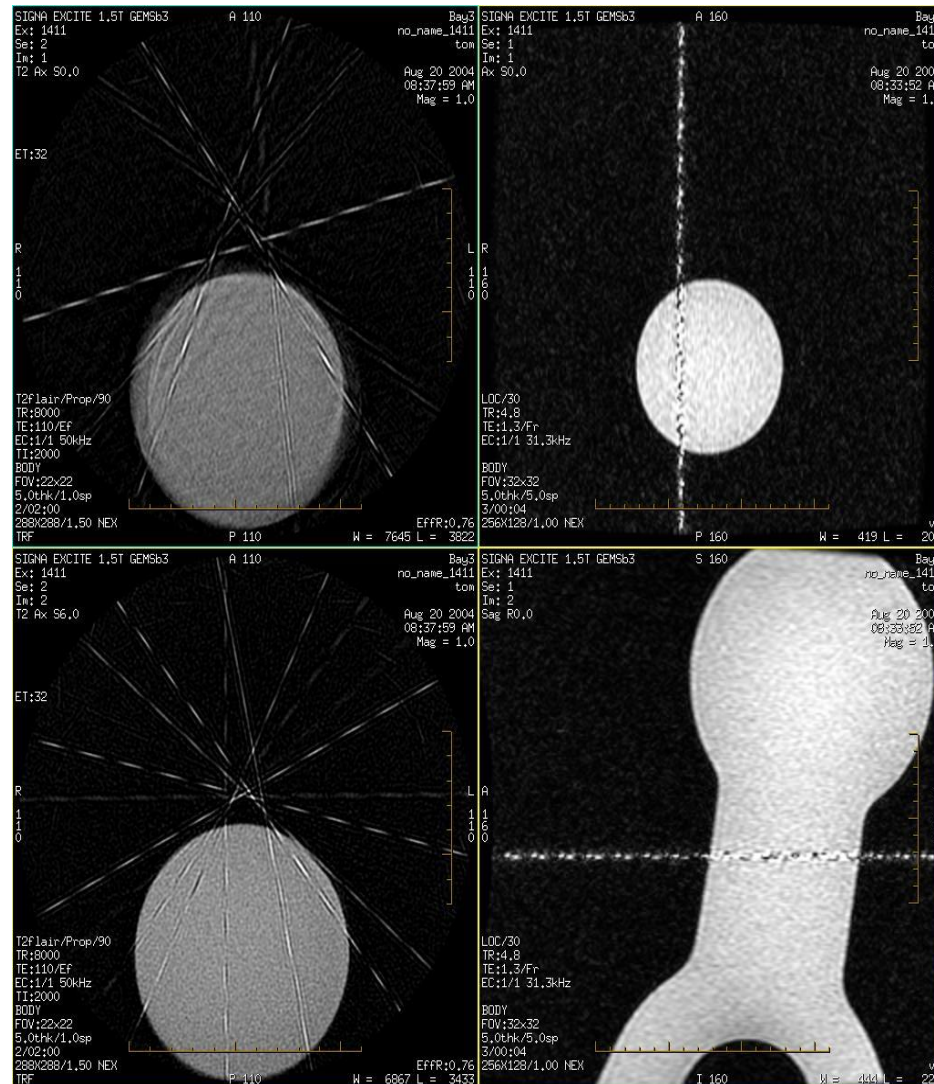


DW EPI...
“corduroy”

RF Leak looks different with Propeller

Propeller...

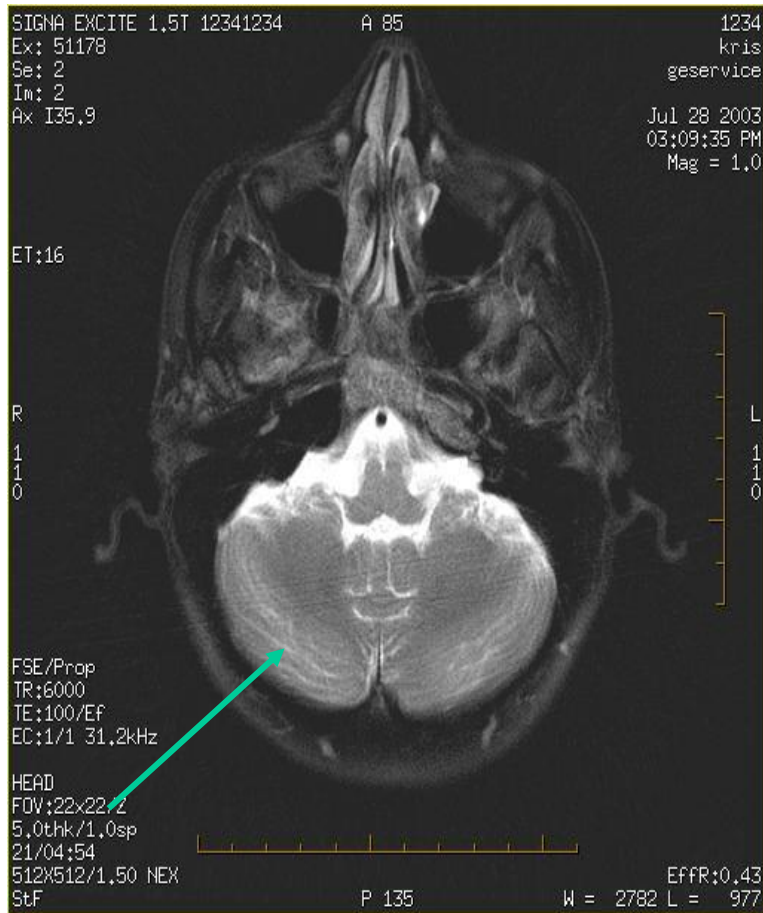
“criss-crossed
lines”



GRE...

“zipper”

Lines criss-crossing the image



- **Issue Description:**

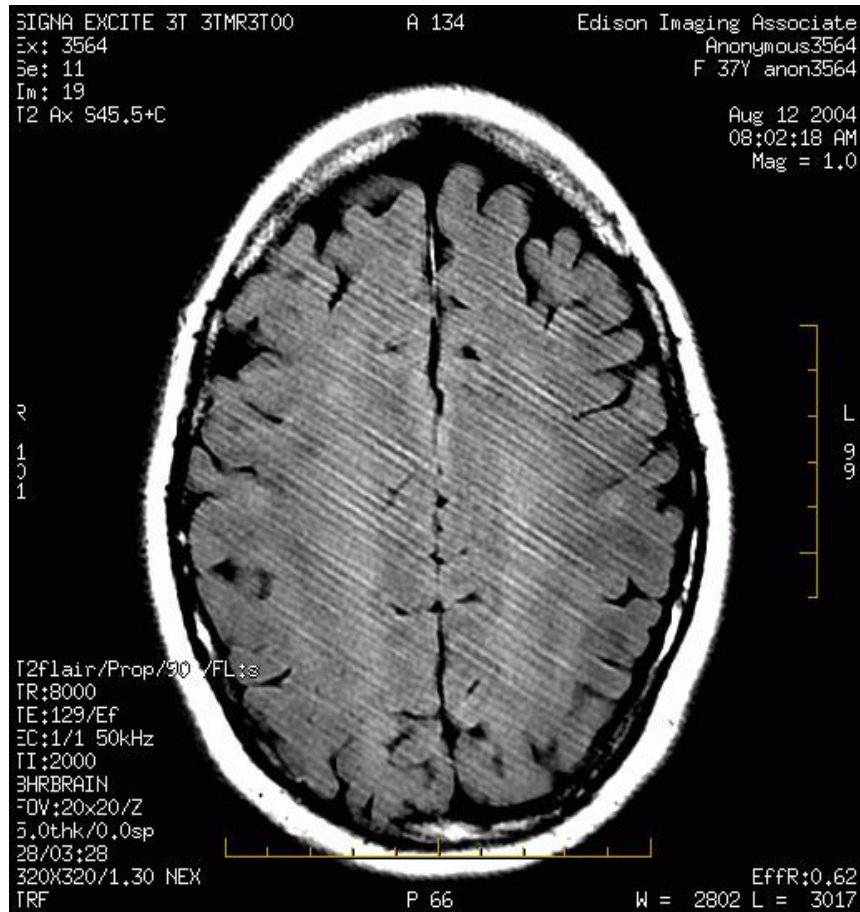
- Lines criss-crossing the image
- Other applications may show a different effect

- **Solution:**

- In this case, the scan room door was open
- If the problem occurs consistently, FE needs to troubleshoot the issue

Affects: All Propeller Applications

Lines criss-crossing the image...another example



- **Issue Description:**
 - Lines criss-crossing the image
- **Solution:**
 - >In this case, possibly caused by in vivo patient monitoring device
 - >If the problem is not patient specific, FE needs to troubleshoot the issue

Affects: All Propeller Applications

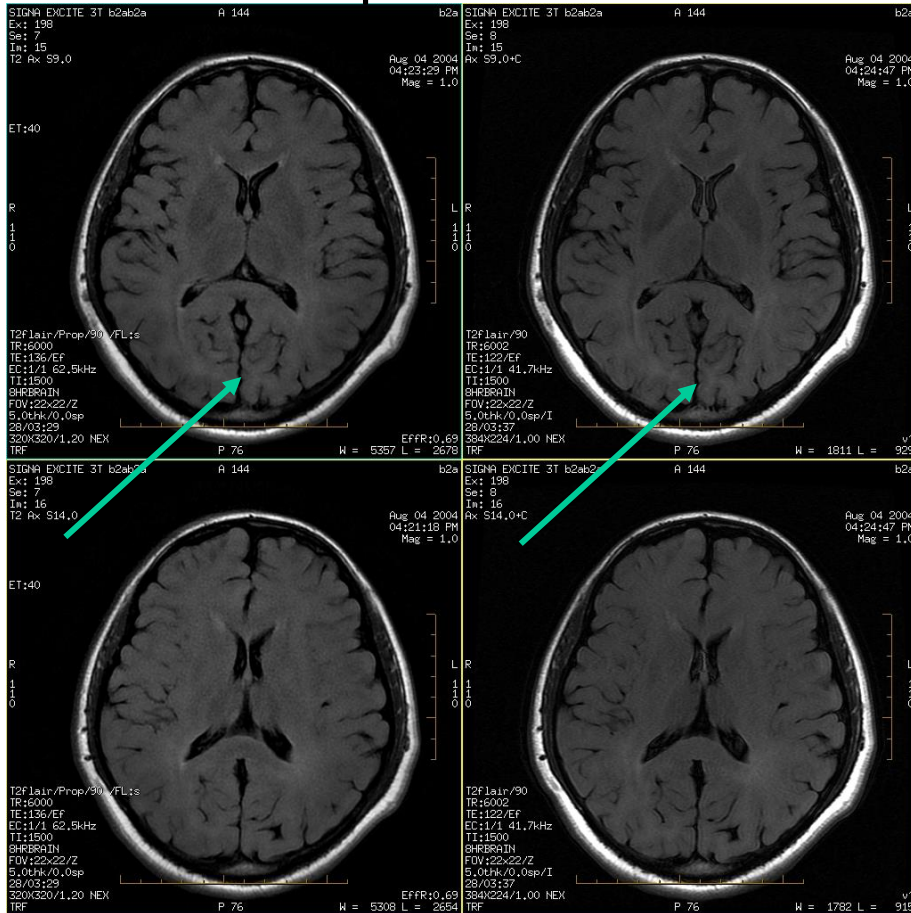
Contrast Issue with low TR in Research mode

- Seen in T2 FLAIR Propeller
- Caused by using a TR below the clinical minimum which results in a very low TI...recall that TI is automatically calculated ($TR/4$)

Contrast Issue from low TR in research mode

T2 FLAIR Propeller

T2 FLAIR



- **Issue Description:**

- Seen when using TRs below the clinical minimum of 8000ms in research mode
- Similar problem seen with FLAIR

- **Solution:**

- Modify protocols...TR should be > 8000

Affects: T2 Flair Propeller

Scanning Propeller in Research mode

- In clinical mode, parameters are limited to a range that is designed to help the operator maintain good image quality.
- In research mode, only system limits are imposed. All coils are allowed. Research mode allows for more freedom of prescription...be careful!



Thank You